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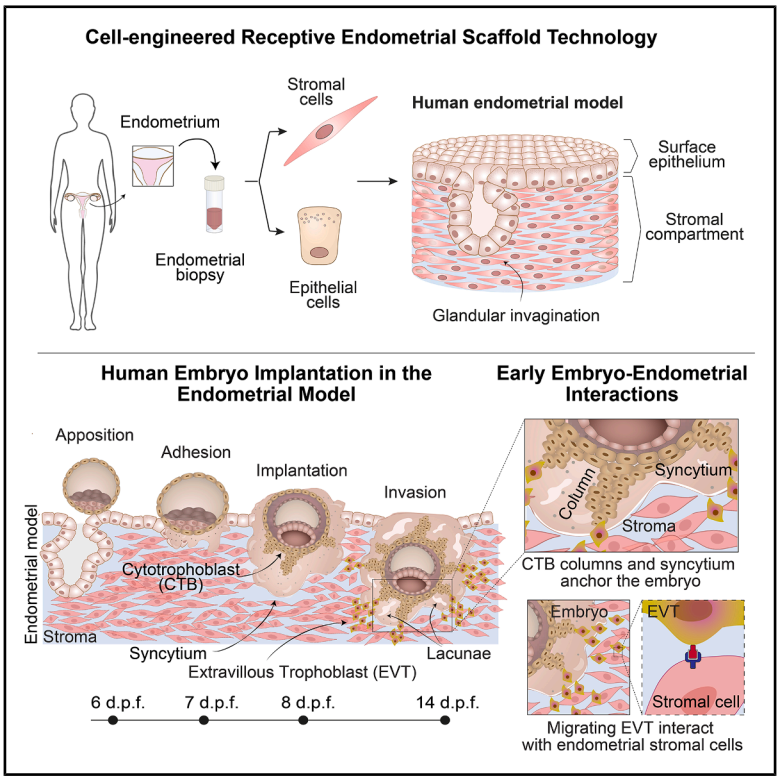
Modeling human embryo implantation *in vitro*

Graphical abstract



Highlights

- Established a 3D *in vitro* model of the receptive human endometrium
- Endometrial model supports *in vitro* human embryo implantation and development
- Advanced development of embryonic and extraembryonic lineages
- Identified ligand-receptor interactions at the embryo-endometrial interface

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In brief

Building an *in vitro* model of receptive endometrium allows investigation of human embryo implantation and early post-implantation development and recapitulates the earliest stages in establishing the human placenta and formation of the embryo-endometrial interface.

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Modeling human embryo implantation *in vitro*

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SUMMARY

Implantation of a human embryo into the endometrium is a crucial event in gestation, as it marks the initiation of a pregnancy and is prone to high failure rates. We have limited understanding of these stages because of the inaccessibility of implanting embryos and the lack of suitable model systems. Here, we establish an *in vitro* model that recapitulates the luminal, glandular, and stromal compartments of the superficial layer of receptive human endometrium. Human embryos and blastoids implant into the endometrial model, achieving post-implantation hallmarks including advanced trophoblast structures that underlie early events in placental development. Single-cell RNA sequencing of the embryo-endometrial interface at day 14 uncovers predicted molecular interactions between conceptus and endometrium. Disrupting signaling interactions between extravillous trophoblast and endometrial stromal cells caused defects in trophoblast outgrowth, demonstrating the importance of crosstalk processes to sustain embryogenesis. This platform opens the opportunity to investigate early stages of human embryo implantation.

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INTRODUCTION

Implantation marks the initiation of pregnancy by establishing connections between the embryo and maternal uterus that are necessary for long-term intrauterine gestation.[1] This process begins between the first and second week following fertilization, when the embryo adheres to the luminal epithelium of the endometrium that lines the uterine cavity. Following attachment, the embryo breaches the epithelial layer and begins invading the underlying decidualized stroma. Glandular cells secrete factors that sustain the development and growth of the embryo.[2,3] During this initial invasion through the superficial layer of the endometrium, trophoblast cells of the conceptus proliferate and differentiate, infiltrating into the decidua and laying the building blocks for placental development.[4] Interactions between embryo and endometrium are tightly regulated, and the implantation niche involves synchronous signals between multiple cell types of the developing embryo and the decidualized endometrium.[1]

These stages of embryo implantation and early post-implantation development have significant clinical relevance.[5–7] The high incidence of implantation failure represents a limiting factor for the success of *in vitro* fertilization (IVF) procedures.[1,8] Additionally, complications including preeclampsia, intrauterine growth restriction, and preterm birth, which manifest later in pregnancy, may originate from defective trophoblast-endometrial interactions established as early as implantation.[9] Understanding the processes underpinning these events is therefore a prerequisite for defining the steps necessary to establish a healthy pregnancy.

Despite its clinical relevance, our knowledge of implantation is limited, partly due to the inaccessibility of this stage as the embryo becomes progressively hidden inside the decidua. As a result, most observations have been derived from examinations of early pregnancy hysterectomy specimens.[10–14] While these studies provide information about the anatomical architecture of the implantation site, they are necessarily static morphological snapshots that preclude dynamic or interventional investigation. To address this gap, *in vitro* models using endometrial cell lines, organoids, and assembloids have contributed valuable insights into how human embryos attach to endometrium.[15–22] However, these models do not yet fully capture the tissue organization and physiological characteristics of receptive endometrium, limiting their ability to model the complete process of implantation and early post-implantation development. Similarly, improved culture conditions have extended the development of human embryos *in vitro*, but these embryos are typically grown in synthetic environments without endometrial tissue, restricting the study of embryo-endometrial interactions.[23–28] Furthermore, existing embryo culture systems show limited trophoblast development, perhaps due to the lack of endometrial cells in current models.

A further challenge is the difference in implantation strategies between humans and other mammalian species, including primates, which vary in the location and invasiveness of the embryo relative to the endometrium, as well as early placentation events.[29,30] This hinders the applicability of results derived from nonhuman model systems. To address this, stem cell-derived blastoids are a recent alternative human embryo model that offers benefits in terms of scalability and accessibility to study embryo implantation.[31–33] Blastoids recapitulate certain developmental features observed in embryos; however, the extent to which they accurately mimic human embryogenesis, and where they diverge, remains incompletely understood.[34,35]

Due to these obstacles, the earliest stages of post-implantation human embryo development within endometrial tissue remain challenging to model *in vitro*, restricting our ability to explore cellular and molecular mechanisms underlying this critical phase. To overcome this barrier, we provide a cell-engineered receptive endometrial scaffold technology (CREST) that models both the epithelial and stromal compartments of the superficial layer of receptive human endometrium, the tissue that receives the implanting embryo. This model supports *in vitro* implantation and early post-implantation development of human embryos and blastoids, enabling visualization and characterization of developmental milestones and early embryo-endometrial interactions.

RESULTS

Modeling cellular organization of human endometrium

To address the limited availability of *in vitro* systems that model the diverse cell types and compartments of the human endometrium,[36] a key barrier to studying embryo implantation, we first isolated epithelial and stromal cells from endometrial biopsies of healthy, fertile donors (Figure 1A; Table S1). Fibroblast-like cells, which populate the stromal layer of endometrium, were isolated and expanded (Figure 1B).[37] Epithelial cells were propagated as organoids, which contain progenitor cells with the potential to differentiate into luminal and glandular cell types[38–41] (Figure 1C).

To build the three-dimensional (3D) endometrial model, stromal cells were encapsulated inside a fully defined hydrogel composed of dextran polymers and cyclodextrin crosslinker and cultured in a transwell insert (Figure 1D). To encourage degradability of the hydrogel during embryo implantation, we chose a cyclodextrin crosslinker containing metalloprotease-cleavable sites. In initial experiments, stromal cells remained largely rounded and proliferated poorly over 4 days of culture within the hydrogel (Figures S1A–S1C). As the crosslinkers do not contain functional sites for cell adhesion, we reasoned that adding extracellular matrix (ECM) proteins that are present in endometrium would support stromal growth.[42] To inform selection of ECM proteins, we used transcriptome data of endometrial biopsies collected during the mid-secretory phase of the menstrual cycle and examined expression levels of ECM constituents (Figure S1D). From among highly expressed factors, we selected collagen I, III, and VI and fibronectin as structural components that can form complex fibrillary networks and incorporated these ECM proteins into the hydrogel (Figure 1D). These conditions enhanced stromal cell proliferation and promoted cell elongation together with the formation of an ordered, interconnected fibroblast-like network throughout the hydrogel (Figures 1E and S1A–S1E).

To construct the endometrial epithelium, we seeded and cultured fragments of endometrial organoids on top of the hydrogel containing stromal cells (Figure 1D). Fragments grew out

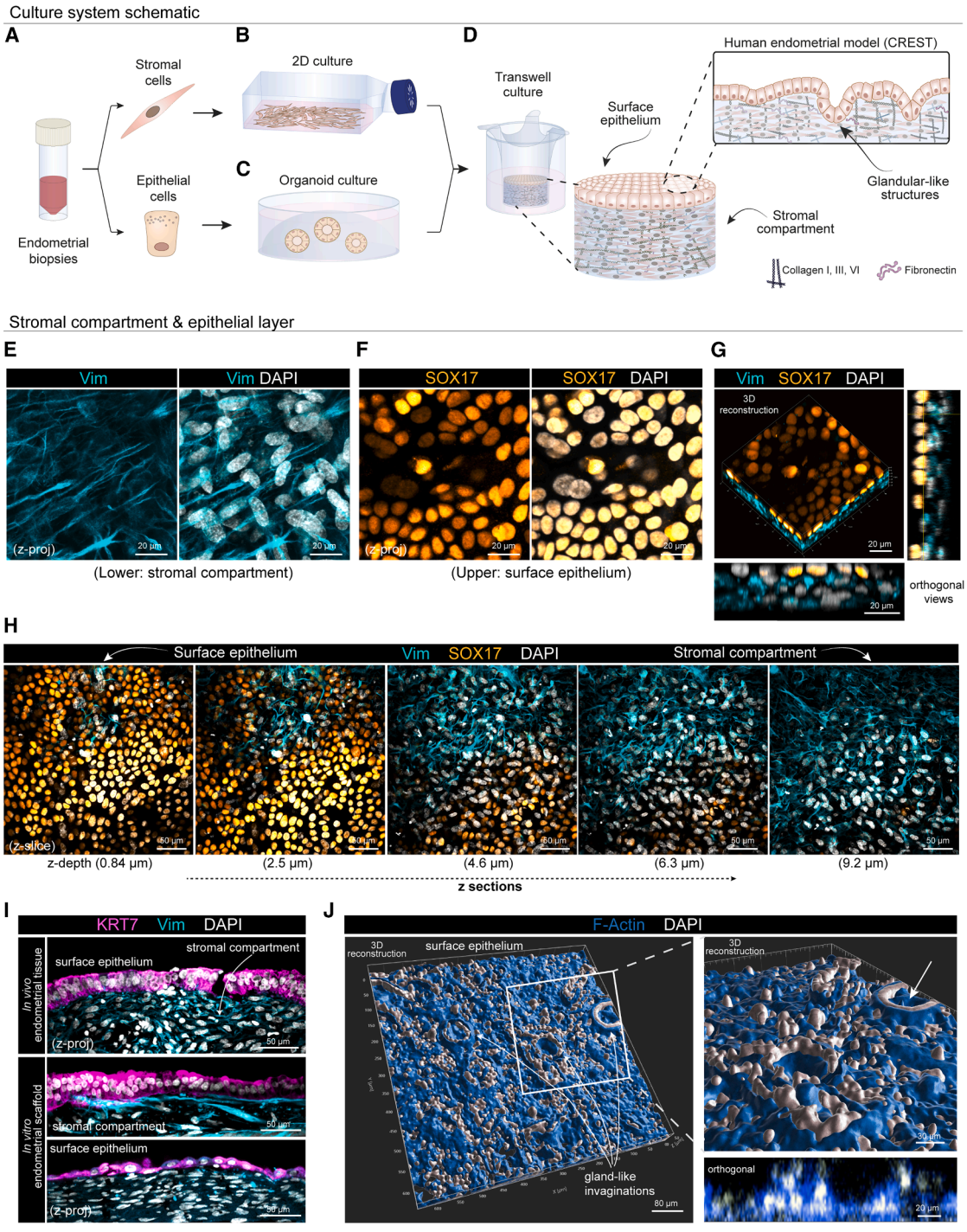

Figure 1. Building a three-dimensional endometrial scaffold

(A) Endometrial stromal and epithelial cells isolated from biopsies.

(B) Stromal cells grown as monolayers.

(C) Epithelial cells propagated as organoids.

(D) Assembly of CREST.

(E) Stromal cells within hydrogel form a compartment of interconnected cells.

(F) Surface epithelium labeled for SOX17.

(G) 3D reconstruction and orthogonal views show the interface between a single epithelial layer and the underlying stromal compartment.

(H) Series of confocal z-sections at increasing depth from the surface of the scaffold.

(legend continued on next page)

within 4 days to form a confluent SOX17-positive epithelial monolayer (Figure 1F). As the culture progressed, epithelial and stromal cells formed an organized, multi-layered tissue (Figures 1G and 1H), which on average was ~180 μm thick with a volume of ~6 mm^3 (Figure S1F) and approximated the stiffness of the *in vivo* endometrium[43] (Figures S1G–S1I). Additionally, the cytoarchitecture and marker expression of the *in vitro* scaffold mirrored *in vivo* endometrial tissue, characterized by a continuous surface epithelium and well-defined underlying stromal compartment as visualized by cross-sectioning (Figure 1I). Small cavities also appeared on the epithelial surface and formed openings, suggesting initiation of gland-like structures (Figure 1J). Altogether, we engineered an *in vitro* model using endometrial biopsy-derived cells that recapitulates key anatomical features of the superficial layer of the human endometrium.

CREST is responsive to hormones and displays features of tissue receptivity

Human embryo implantation occurs during a brief window in the mid-secretory phase of the menstrual cycle,[44] when hormonal signals from estrogen (E2) and progesterone (P4) render the uterine tissue receptive.[45] To recreate this window, stromal and epithelial compartments of the endometrial scaffold were hormonally stimulated for 4 days with E2 to mirror the natural peak of E2 during the proliferative phase, followed by E2, P4, and cyclic AMP (cAMP) for 8 days to induce differentiation toward the mid-secretory phase, following previously established regimes.[39,40] Upon hormone stimulation, stromal cells underwent cytoskeletal reorganization from elongated to cuboidal morphologies (Figures 2A and S1J). Hormonal stimulation also induced the acquisition of a secretory-like phenotype in stromal cells, evidenced by increased secretion of prolactin (PRL) (Figure 2B) and insulin-like growth factor binding protein 1 (IGFBP1) (Figure 2C). Acquiring these secretory capabilities together with morphological changes are hallmarks of endometrial decidualization during the mid-secretory phase.[45]

Epithelial cells also underwent morphological and functional remodeling during this process. Cells on the epithelial surface became less round and displayed irregular borders (Figures S1K–S1M). To characterize epithelial cell types generated *in vitro*, we assessed markers of ciliated (α-tubulin-positive cilia) and glandular (FOXA2-positive nuclei) cell types over hormonal stimulation, as these lineages represent functional components of endometrial epithelium.[46,47] Initially, epithelial organoid fragments seeded onto scaffolds lacked cilia and glandular markers (Figure S1N). However, multiciliated cells were then observed on ~50% of surface epithelium during the proliferative phase (Figures 2D, 2E, and S1O). In the secretory phase, the proportion of ciliated cells decreased compared with the proliferative phase (Figures 2D and 2E), a change consistent with *in vivo* data.[47–50] Conversely, we observed a 5-fold higher proportion of FOXA2-positive cells in the secretory phase compared with the proliferative phase (Figures 2D–2F), indicating a shift toward secretory identity. Notably, we also observed pinopodes on the apical surface of the epithelium (Figure 2G). These bleb-like structures *in vivo* are limited to the window of implantation, represent a hallmark of endometrial receptivity,[51,52] and may act as sites of anchorage for embryo apposition and adhesion.[52,53]

To further investigate these morphological changes, we performed live-cell imaging over different phases of hormonal stimulation. We focused on the glandular-like openings first observed in unstimulated epithelium and tracked these same regions through proliferative and secretory phases (Figures 2H, 2I, and S2A). After 4 days of E2 stimulation, glandular openings deepened (Figures 2H and 2I), and tube-like structures formed, protruding toward the underlying stromal compartment (Figures 2I and S2B). These changes mirror the proliferative phase *in vivo*, during which E2 promotes endometrial proliferation and early stages of glandular development. Transition to P4 induced spontaneous formation of cup-like structures and branched tubular networks across and underneath the epithelial layer (Figures 2I, S2C, and S2D). These features align with the secretory phase of the *in vivo* endometrium, where P4 promotes glandular maturation and reorganization.[54]

Maturation of glandular structures was also accompanied by a gain in functional hallmarks. Levels of glycodelin (PAEP), a glycoprotein enriched in glandular secretions, increased progressively in the culture medium following hormonal stimulation (Figure 2J). Direct access to these glandular secretions, collectively known as "uterine milk,"[2,55] is essential for histotrophic nourishment of the conceptus during the first few weeks of pregnancy.[2,55] In contrast to previous endometrial organoid models,[20,56] where secretions are often trapped inside the lumen,[57] glandular epithelium in our endometrial scaffold released secretions directly into the surrounding tissues. Notably, cells producing glycodelin clustered at the border of glandular-like openings (Figures 2K and S1P). Importantly, we confirmed that endometrial cells retained viability after scaffold construction and hormone-induced differentiation (Figure S1Q).

Altogether, these results demonstrate that CREST reproduces morphological features of endometrial organization and functional responsiveness to hormonal stimulation. Importantly, the model displays characteristics of endometrial receptivity reminiscent of the window of implantation during the mid-secretory phase of the *in vivo* menstrual cycle.

Blastoids undergo adhesion and implantation into CREST

Previous studies have shown blastoids attach to endometrial monolayers[31] and assembloids,[21] indicating compatibility for the initial attachment phase of implantation *in vitro*. However,

(I) Cross-section of *in vivo* endometrial biopsy and *in vitro* endometrial CREST scaffolds.
(J) 3D reconstruction of the CREST epithelial surface shows gland-like invaginations. Scaffolds were labeled with live-cell fluorescent dyes for nuclei and actin. Images are representative of two (I and J), three (F, G, and H), or five (E) independent experiments. Scale bars, 20 μm (E, F, G, and J insert), 30 μm (J), 50 μm (H and I), and 80 μm (J).
See also Figure S1 and Table S1.

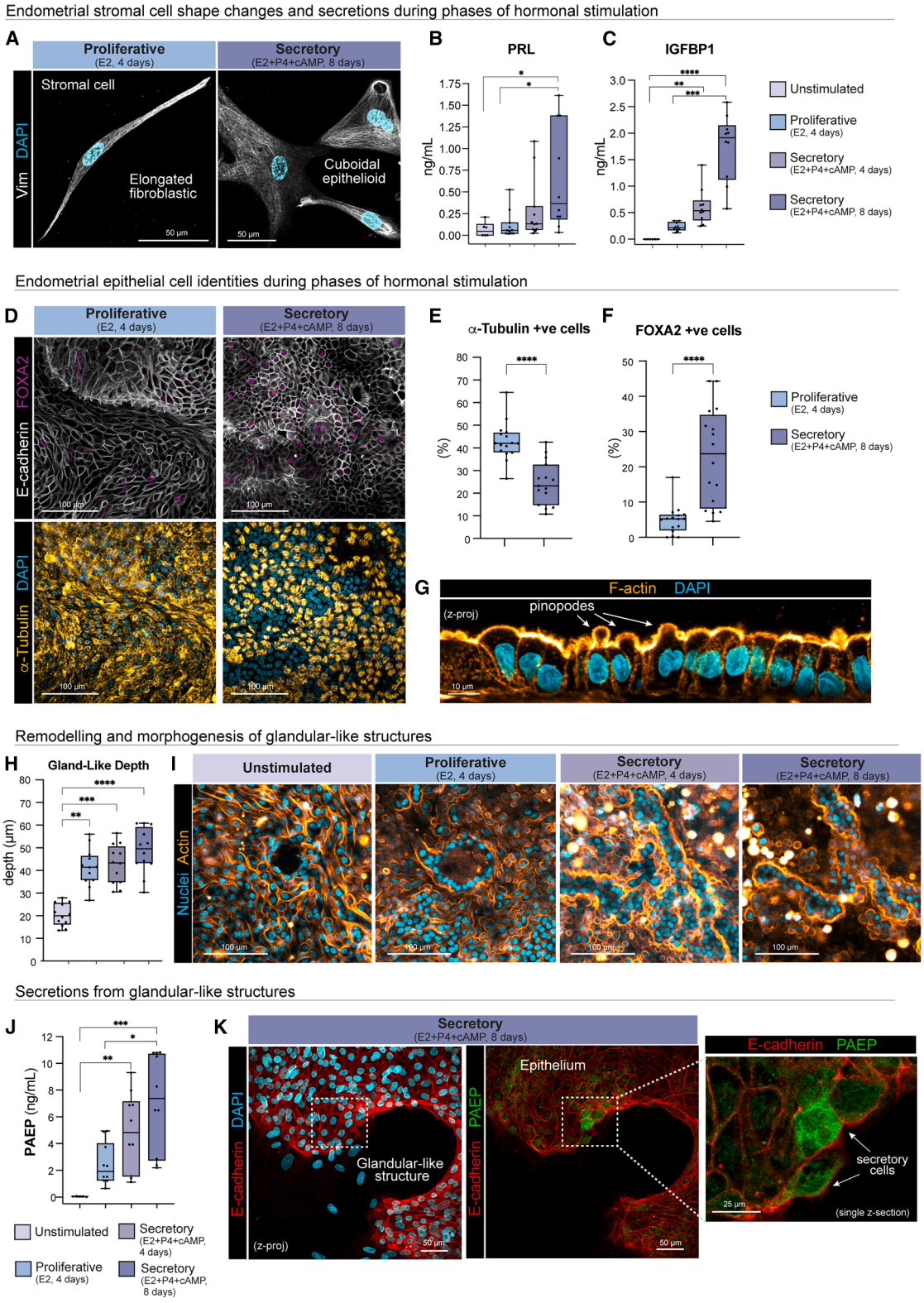

Figure 2. Endometrial scaffolds respond to hormonal stimulation and acquire features of endometrial tissue receptivity

(A) Endometrial stromal cells display an elongated appearance during the proliferative phase (E2) and cuboidal morphology in the secretory phase (E2, P4, and cAMP).

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whether blastoids can model subsequent implantation stages, including tissue invasion, remains unknown.

To address this gap, we generated blastoids from human naive embryonic stem cells[33] (blastocyst-derived HNES1 line[58]) and transferred them to endometrial scaffolds, assessing their ability to undergo implantation. After 24 h, 53% of blastoids had attached or implanted, with attachment defined by firm adhesion of the expanded blastocyst-like structures to the epithelium (Figures S3A–S3C). This process peaked at 72 h with 86% of blastoids attached or implanted (Figure S3B).

Following implantation, we next assessed whether CREST could sustain blastoids through early post-implantation stages. For human embryos, 6 days of post-implantation culture corresponds approximately to 12 days post-fertilization (d.p.f.), and by this time blastoids exhibited hallmarks of this stage, including epiblast-like and hypoblast-like cells, expanded amniotic and yolk sac cavities (Figures S3D–S3G), and proliferation and differentiation of trophoblast subtypes such as extravillous-like and syncytiotrophoblast (STB)-like cells (Figures S3H and S3I). While the organization of embryonic and extraembryonic structures was variable within this limited series, blastoids were able to undergo adhesion and implantation and form post-implantation lineages, thereby establishing that CREST supports blastoid implantation.

Implantation of human embryos in CREST

We next tested the capacity of CREST to sustain implantation of human embryos (Figure 3A). To this end, human blastocysts at 5 d.p.f. were transferred into transwells containing hormone-stimulated endometrial scaffolds (Figure 3B). Following embryo transfer, stromal cells (lower compartment) were cultured in medium containing 2% charcoal-stripped (CS) fetal bovine serum (FBS) supplemented with E2 and P4. By contrast, the epithelial layer and pre-implantation embryo in the upper compartment were maintained in a distinct medium containing 2% CS FBS, 5% human serum albumin, N2, B27, antioxidants, E2, and P4, named as Human Embryo Medium 1 (HEM 1).

Over the next 24 h (5–6 d.p.f.), blastocysts hatched from the zona pellucida and expanded their blastocoel cavity (Figure 3B). By ~6–7 d.p.f., embryos initiated loose contact with the epithelial surface, orienting the polar trophectoderm toward the scaffold, a process known as apposition. By ~7 d.p.f., stronger attachment was evident, with the blastocoel still inflated, marking the transition to adhesion, defined by loss of movement during gentle agitation of the plate. Implantation typically occurred by 8 d.p.f. (Figures 3C and 3D), characterized by deflation of the blastocoel and acquisition of a dome-shaped morphology (Figure 3B). By 10 d.p.f., embryos had penetrated the epithelial layer and were no longer visible on the surface. Implantation was achieved in 60% of transferred embryos (n = 52/86; Figure 3E), confirming that the CREST platform supports human embryo implantation *in vitro*.

We next leveraged CREST to profile proteins secreted at the embryo-endometrial interface. These secretions sustain early pregnancy; however, the timing, duration, and relative levels of these secretions in early post-implantation embryo development have been difficult to study. To address this gap, we first measured human chorionic gonadotropin (hCG), a biochemical marker used in the recognition of clinical pregnancies. hCG levels secreted by embryos at 6 d.p.f. in our cultures (Figure S3J) were similar to previous reports at the same stage of embryo development.[59] Following implantation into CREST, hCG secretion increased over the following days, indicating development of the trophoblast compartment (Figures 3F and S3J). We then measured other pregnancy-associated protein secretions, including human placental lactogen (hPL), human pregnancy-associated plasma protein-A (PAPP-A), and placental growth factor (PlGF). hPL levels remained stable from 7 to 11 d.p.f., followed by an increase from 12 to 14 d.p.f. (Figures 3F and S3J). By contrast, PAPP-A and PlGF levels showed a sharp increase at 10 d.p.f., continuing with a gradual increase to 14 d.p.f. (Figures 3F and S3J). Collectively, these results provide insights into the timing and dynamics of protein secretion following implantation and highlight the complex

(B) Increased PRL secretion detected by enzyme-linked immunosorbent assay (ELISA) at 8 days of the secretory phase compared with the unstimulated (*$p < 0.05$) and proliferative phases (*$p < 0.05$): Kruskal-Wallis test corrected by Dunn's (6–11 independent experiments).
(C) Increased IGFBP1 secretion measured by ELISA at 8 days of the secretory phase compared with the unstimulated (****$p < 0.0001$) and proliferative phase (***$p < 0.001$), and at 4 days of the secretory phase compared with the unstimulated (**$p < 0.01$): Kruskal-Wallis test corrected by Dunn's (6–10 independent experiments).
(D) Top views of CREST epithelium in proliferative (E2) and secretory (E2, P4, and cAMP) phases show glandular epithelial cells (FOXA2), ciliated cells (acetylated α-tubulin), epithelial junctions (E-cadherin), and nuclei (DAPI).
(E and F) Percentage of (E) ciliated cells and (F) FOXA2-positive cells out of total nuclei (DAPI-positive). A total of 16 regions per time point were quantified, pooled across replicates (2 independent experiments for proliferative, 4 for secretory). Values were compared by the unpaired one-way Mann-Whitney test, corrected by Dunn's. ****$p < 0.0001$.
(G) Cross-section of hormone-stimulated surface epithelium shows pinopodes.
(H) Quantification of gland-like invagination depth in surface epithelium over the course of hormonal stimulation. Depth increased from unstimulated to proliferative (**$p < 0.01$), 4-day secretory
(***$p < 0.001$), and 8-day secretory phases (****$p < 0.0001$): Kruskal-Wallis test corrected by Dunn's (n = 1 independent experiment).
(I) Live imaging of the same CREST scaffold region across stimulation stages, with live-cell fluorescent labeling of nuclei and actin.
(J) Increased PAEP levels measured by ELISA at 8 days of the secretory phase compared with unstimulated (***$p < 0.001$), proliferative phase (*$p < 0.05$), and at 4 days of the secretory phase compared with unstimulated (**$p < 0.01$): Kruskal-Wallis test corrected by Dunn's (5–10 independent experiments).
(K) PAEP in subpopulation of epithelial cells lining cavities.
Plots in (B), (C), (E), (F), (H), and (J) show the median, horizontal line; interquartile range, boxes; and min and max values, whiskers. Images labeled z-proj display the max projection of z stack, and z-section represents a single plane. Images are representative of two (I) or three (A, D, and K) independent experiments. Scale bars, 10 μm (G), 25 μm (K expanded), 50 μm (A, D, and K main), and 100 μm (D and I).
See also Figures S1 and S2.

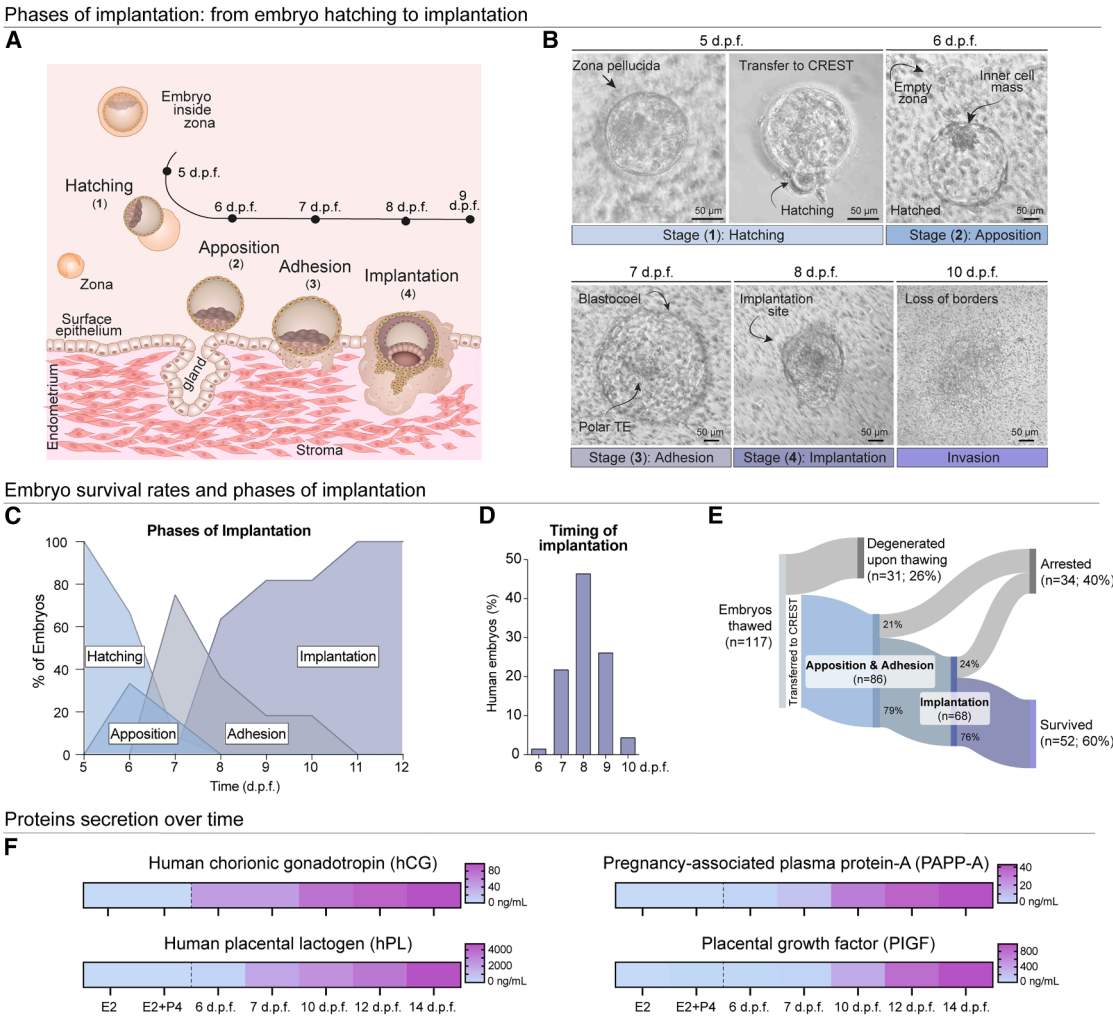


Figure 3. Implantation of human embryos *in vitro*

(A) Schematic of experiment and stages of human embryo development *in vitro*.

(B) Human embryos at 5 d.p.f. were thawed and transferred into CREST scaffolds. Bright-field images show embryos during hatching (Stage 1), apposition and adhesion (Stages 2 and 3), and implantation and invasion (Stage 4). Scale bar, 50 µm.

(C) Progression of embryos through implantation phases over time. Only embryos that successfully reached implantation were included in this analysis.

(D) Frequency of embryo implantation across d.p.f.

(E) Percentages of successfully thawed and transferred embryos undergoing apposition and adhesion (Stages 2 and 3); implanting successfully and growing until the endpoint of the culture (Stage 4). Number of embryos assessed at each stage is shown.

(F) Measurement of secreted hCG, hPL, PAPP-A, and PIGF at different stages by ELISA. E2 and E2 + P4 refer to the proliferative and secretory phases of the endometrial scaffold, respectively. Dashed vertical line indicates transfer of embryos at 5 d.p.f. to endometrial scaffold. Time points 6 to 14 d.p.f. refer to the day of embryo development. Data show means from three independent experiments.

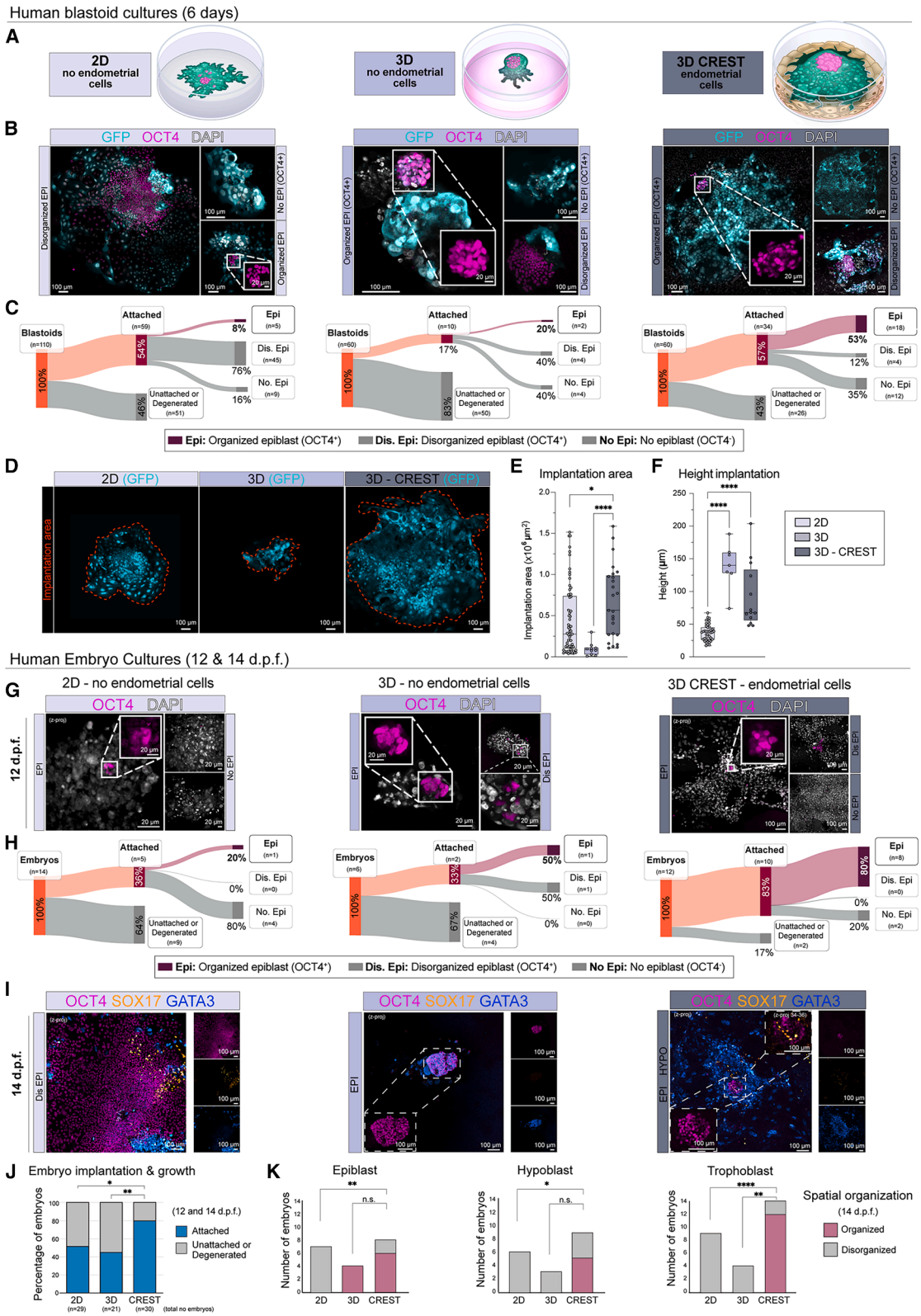
See also Figure S3.

embryo-endometrium interactions that drive early post-implantation development.

Endometrial cells promote organized embryo development

Having established that CREST supports implantation, we next examined how spatial architecture and endometrial cell signaling contribute to embryo development. To investigate this, we compared embryo development across the following systems: (1) a "2D" embryo attachment model, which allowed embryos to spread superficially but not invade; (2) a "3D" hydrogel supplemented with ECM proteins (the same structural components as in CREST) but devoid of endometrial cells, which allowed embryos to spread and invade in 3D; and (3) our 3D CREST system containing endometrial epithelial and stromal cells to directly examine the role of endometrial cells on embryo development (Figure 4A).

To benchmark how these environments support patterning independently of embryonic heterogeneity, we first used blastoids as a reference model to assess rates of succesful



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implantation and embryo survival under different culture conditions. Compared with the 3D hydrogel-only condition, the 2D and CREST conditions supported blastoid attachment and continued development at higher rates (2D: 54%, n = 59/110; 3D: 17%, n = 10/60; CREST: 57%, n = 34/60; Fisher's exact test: CREST vs. 3D, $p < 0.0001$) (Figures 4B and 4C). However, despite comparable attachment rates, CREST promoted a higher proportion of blastoids with organized epiblast-like development than the 2D condition (2D: 8%, n = 5/59; 3D: 20%, n = 2/10; CREST: 53%, n = 18/34; Fisher-Freeman-Halton test: CREST vs. 3D, $p < 0.0001$) (Figures 4B and 4C) and more extensive outgrowth compared with the 2D and 3D conditions (Figures 4D–4F). Thus, while 3D hydrogels without endometrial cells permit epiblast organization, this occurs with reduced efficiency and limited outgrowth, whereas CREST achieves both high attachment rates and organized epiblast development.

As the extent to which blastoids model human embryonic development remains unclear, we next tested whether implantation dynamics and developmental features observed in blastoids translated to embryos in the same three conditions. Consistent with blastoid findings, embryos in CREST showed higher rates of attachment and continued development compared with the 3D condition (80%, n = 24/30 vs. 43%, n = 9/21; Fisher's exact test, $p < 0.01$) (Figures 4G–4J). However, unlike the blastoid model, this difference was also significant when compared with the 2D condition (80%, n = 24/30 vs. 52%, n = 15/29; Fisher's exact test, $p < 0.05$) (Figure 4J). Moreover, in line with blastoid findings, CREST tended to promote a higher proportion of attached embryos with organized epiblast development at 12 d.p.f. relative to 2D conditions (CREST 80%, n = 8/10 vs. 2D 20%, n = 1/5. Fisher-Freeman-Halton test: CREST vs. 2D, $p < 0.05$) (Figures 4G and 4H).

Given the enhanced epiblast organization in CREST, we asked whether this extended to other embryonic lineages: hypoblast and trophoblast. We extended the culture of embryos to 14 d.p.f. and assessed epiblast (OCT4), hypoblast (SOX17), and trophoblast (GATA3 or TFAP2C) populations. While embryos in CREST (50%, n = 7/14) and 2D (56%, n = 5/9) conditions showed higher frequency of lineage marker expression compared with the 3D condition (20%, n = 1/5) (Figure S4A), striking differences emerged in spatial organization. In particular, embryos in 2D culture lacked spatial organization, growing as flat expanses of predominantly OCT4-positive cells, with none displaying morphologically organized lineages (Figures 4I, 4K, S4B, and S4C). By contrast, embryos in 3D hydrogel-only culture exhibited organized epiblast morphology but showed minimal hypoblast and trophoblast development and little outgrowth beyond the central epiblast mass (Figures 4I, 4K, S4B, and S4C). Notably, embryos in CREST exhibited the highest degree of spatial organization, with clear compartmentalization of all three lineages (epiblast: 75%, n = 6/8; hypoblast: 56%, n = 5/9; trophoblast: 86%, n = 12/14) (Figures 4I, 4K, S4B, and S4C).

These findings show that endometrial cells promote attachment and post-implantation development of embryos and blastoids at higher efficiency and with improved spatial organization of primary lineages. This suggests that endometrial cells in CREST provide a supportive physical and biochemical environment to sustain embryo growth *in vitro*.

CREST supports development and organization of embryo lineages post-implantation

Building on the observation that CREST supports embryonic development with improved spatial organization, we next examined in greater depth their development using immunofluorescence microscopy at multiple stages. By 10 d.p.f., embryos reached several milestones, including formation of a yolk sac cavity demarcated by flat, dispersed SOX17+ endoderm cells (Figures 5A and S4D). The yolk sac is thought to be the first extra-embryonic structure to form and will play essential roles in nutrient exchange, hematopoiesis, and patterning the embryo.

Figure 4. Comparison of *in vitro* embryo culture methods

(A) Schematic of *in vitro* culture methods: 2D culture, 3D culture in hydrogel only without endometrial cells, and 3D CREST culture in hydrogel containing epithelial and stromal endometrial cells.

(B) Epiblast characterization of blastoids generated from GFP-expressing HNES1 cells and cultured for 6 days post-formation under the three culture conditions. Epiblast identity was categorized into (1) no OCT4 expression, (2) disorganized OCT4 expression, and (3) organized OCT4 expression.

(C) Quantification of survival of blastoids at the end of culture categorized based on attachment to substrate. Attached blastoids were immunostained with DAPI and OCT4 to assess epiblast organization (two independent experiments). Fisher-Freeman-Halton test was used to compare organized epiblast, disorganized epiblast, and no epiblast: CREST vs. 2D ($p < 0.0001$); CREST vs. 3D (p = n.s.). Fisher's exact test was used to compare attached vs. unattached/degenerated: CREST vs. 2D (p = n.s.); CREST vs. 3D ($p < 0.0001$).

(D) Implantation site morphology in 2D, 3D, and CREST systems based on maximum projection of GFP. Implantation area, red outline.

(E and F) Quantification of implantation site area (E) and height (F). Values compared by unpaired one-way Kruskal-Wallis test, corrected by Dunn's (*$p < 0.05$, **$p < 0.01$, ****$p < 0.0001$). Horizontal line, median; boxes, interquartile range; whiskers, min and max values (two independent experiments).

(G) Epiblast characterization in 12 d.p.f. human embryos under conditions in (A). Epiblast structure assessed and categorized as in (B) and (C).

(H) Quantification of 12 d.p.f. embryo survival scored for attachment and OCT4-based epiblast organization (two independent experiments). Fisher-Freeman-Halton test was used to compare organized epiblast, disorganized epiblast, and no epiblast: CREST vs. 2D (**$p = 0.0002$); CREST vs. 3D (p = n.s.). Fisher's exact test was used to compare attached vs. unattached/degenerated: CREST vs. 2D (*$p = 0.0292$); CREST vs. 3D (**$p = 0.0086$).

(I) Epiblast, hypoblast, and trophoblast assessed in 14 d.p.f. human embryos using OCT4, SOX17, and GATA3 immunostaining as lineage markers.

(J) Quantification of embryo attachment and growth across conditions including embryos at 12 and 14 d.p.f. Values compared by Fisher's exact test (*$p < 0.05$, **$p < 0.01$) (two independent experiments for 2D and 3D; one independent experiment for CREST).

(K) Quantification of epiblast, hypoblast, and trophoblast organization in 14 d.p.f. embryos. Values compared by Fisher's exact test (*$p < 0.05$, **$p < 0.01$, ****$p < 0.0001$).

Images are representative of one (I, CREST) or two (B, D, G, and I; 2D and 3D) independent experiments. Scale bars, 20 μm (G, expanded panels in B and G), 100 μm (B, D, G, and I).

See also Figure S4.

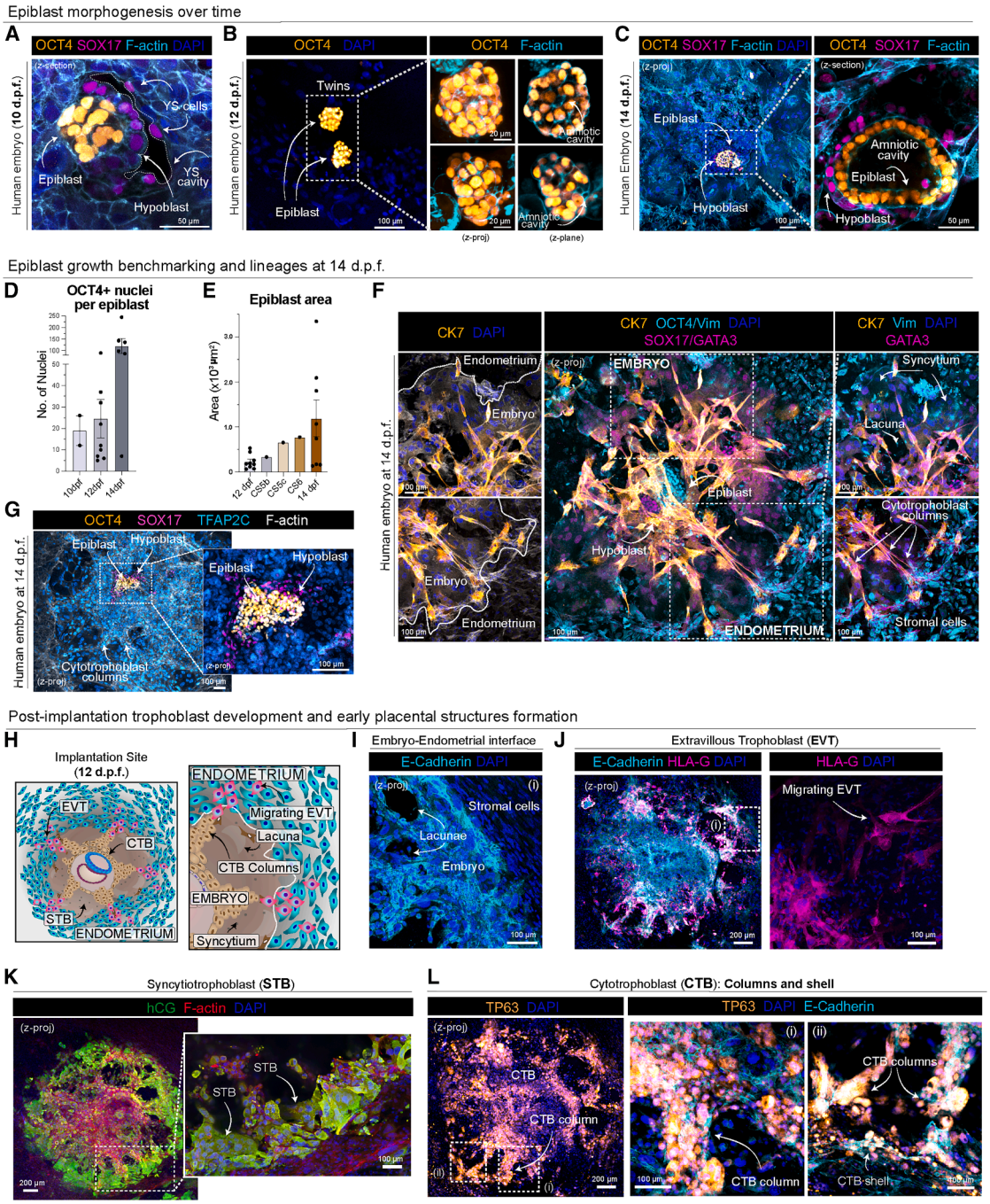

Figure 5. Early post-implantation development of human embryos

(A) 10 d.p.f. implanted human embryo: formation of rosette in epiblast (OCT4+), hypoblast with columnar morphology of SOX17+ cells lining epiblast, and yolk sac cavity demarcated by flat, dispersed SOX17+ cells (2/2 embryos).

(B) 12 d.p.f. implanted human embryo: twinning with initial formation of amniotic cavity (1/2 embryos).

(C) 14 d.p.f. implanted human embryo: formation of epiblast and hypoblast bilaminar disc, amniotic and yolk sac cavities (2/2 embryos).

(D) OCT4+ nuclei per embryo. Data show mean and SEM from 1 (10 d.p.f.) and 3 independent experiments (12 and 14 d.p.f.).

(E) 12 and 14 d.p.f. embryo implantation site area (2 independent experiments) compared with CS5b, CS5c, and CS6 specimens. Data show mean and SEM.

(F) 14 d.p.f. embryo implantation site: embryo marked by epiblast (OCT4+), hypoblast (SOX17+), and trophoblast (GATA3+, KRT7+); surrounding decidualized endometrial stromal cells marked by vimentin (VIM). Inserts: contacts between embryo and endometrial cells.

(G) 14 d.p.f. embryo implantation site: epiblast (OCT4+), hypoblast (SOX17+), and trophoblast (TFAP2C+) (2/2 embryos).

(H) Schematic of images in (F–L).

(I) Embryo-endometrial interface between trophoblast (E-cadherin+) and endometrial stromal cells (E-cadherin−) (2/2 embryos).

(legend continued on next page)

Contrasting this structure, hypoblast cells located along the basal domain of the epiblast retained a columnar and tightly packed morphology (Figures 5A and S4D), indicating the presence of regionally distinct hypoblast derivatives.

To build on these early lineage distinctions, we next examined *in vitro*-implanted embryos at 12 and 14 d.p.f. (Figures 5B and 5C), stages that have remained largely unexplored due to the difficulty of maintaining embryo viability *in vitro* over time. Embryos implanted into CREST showed advanced morphological features, including expansion of the epiblast by 12 d.p.f. (Figure 5B) and formation of a bilaminar disc at 14 d.p.f. (Figures 5C, S4E, and S4F). Cavities flanked the bilaminar disc, consistent with the formation of amniotic and yolk sac cavities (Figures 5C, S4E, and S4F). While identification of the amnion at this developmental stage has historically been restricted by the absence of definitive molecular markers, we observed a population of TFAP2C+, OCT4+, SOX2− cells adjacent to the amniotic cavity, suggestive of putative amniotic lineage (Figure S4G). Interestingly, we also detected putative extraembryonic mesoderm cells overlaying the hypoblast and lining the outer surface of the yolk sac cavity (Figures S4H and S4I). As robust markers for extraembryonic mesoderm are lacking, the marker used here, vimentin (VIM), is expressed in, but not specific to, this lineage. Nevertheless, the location and morphology of these cells are consistent with histological reports.[60,61]

To relate morphological observations to developmental progression, we examined proliferation and expansion of the epiblast compartment. Epiblast cell number progressively increased from 10 to 14 d.p.f. (Figure 5D). The epiblast area also expanded, with measurements of *in vitro* embryos at 12 and 14 d.p.f. similar to *in utero* embryos at Carnegie stages (CS) 5b, 5c, and 6 (Figure 5E). To integrate these growth measurements with broader lineage relationships, we examined 14 d.p.f. embryos using immunofluorescence microscopy, which revealed well-developed features from trophoblast-, epiblast-, and hypoblast-derived tissues (Figures 5F and 5G).

In contrast with the more modest trophoblast development observed in previous culture systems, embryos in CREST displayed extensive trophoblast proliferation and differentiation. Cell populations and structures characteristic of early placental morphogenesis were detected from 12 d.p.f. onward. Notably, the conceptus was embedded into the endometrium with the trophoblast outgrowth in direct contact with stroma (Figures 5F–5I and S4J; Video S1). At the leading edge of invasion, mononuclear CK7-positive cytotrophoblast (CTB) cells are interspersed with multi-nucleated STB arranged in plate-like assemblies (Figure 5F). This syncytium was SDC1-positive and hCG-positive (Figure 5K), confirming its identity as STB, a specialized trophoblast subtype that supports embryo invasion. Within this syncytial mass, we observed fluid-filled lacunae (Figures 5K and 5I), equivalent to those seen *in vivo* from CS5b onward,[10] which subdivided the syncytial layer into a network of trabeculae. *In vivo*, this is a key step in establishing the lacunar stage, during which fluid-filled spaces form and later become filled with maternal blood, creating the interface for maternal-fetal exchange of oxygen and nutrients. Additionally, HLA-G-positive cells, marking extravillous trophoblast (EVT), were detected migrating into the endometrial stroma in CREST-cultured embryos (Figure 5J). These findings indicate that CREST supports advanced differentiation and spatial organization of the trophoblast lineages.

Of particular interest was the appearance of CK7+, TP63+, and E-cadherin+ trophoblast cells clustered into columns indicative of CTB cell identity (Figure 5L). These columns extended through the trabeculae to the margin of the implantation site, similar to the earliest steps of villous tree development described *in vivo*. These columns, which are first seen in CS5c embryos, mark the formation of primary villi, the initial step in establishing placental villous trees. At the interface with endometrial tissue, lateral expansion of CTB columns appears to give rise to the earliest stages of the cytotrophoblastic shell (Figure 5Lii, zoom).[10,62,63] This shell forms a continuous layer between villi and decidua, stabilizing the implantation site and anchoring the conceptus to the uterine wall. Hence, CREST effectively models aspects of the earliest stages of human placental development and formation of the embryo-endometrial interface.

Molecular characterization of the human embryo implantation niche

To investigate molecular features of the early embryo-endometrial interface, we isolated 17 embryo implantation sites within endometrial scaffolds at 14 d.p.f. and performed single-cell RNA sequencing (scRNA-seq) (Figure 6A). To facilitate cell type identification, we integrated our dataset with a composite dataset of *in vitro*-cultured human embryos[28] and of endometrial biopsies[47] (Figure 6A, right). Lineages were annotated based on cluster identity, expression of marker genes, comparison to reference datasets, and SNP genotyping (Figures 6A and 6B). This approach identified cell sub-populations of the conceptus (Figures 6A, 6B, and S5A–S5F) comprising CTB (n = 133), STB (n = 337), EVT (n = 23), hypoblast/visceral endoderm (Hypo, n = 17), yolk sac endoderm (YS, n = 60), extraembryonic mesoderm (ExMes, n = 35), and amniotic epithelium (Am, n = 76). Despite being abundantly detected by microscopy in about half of all *in vitro* implantation sites (Figure 4), unambiguously identifying embryonic disc epiblast cells in the transcriptome dataset was challenging, perhaps due to sensitivity and loss during single-cell dissociation. We curated a set of putative embryonic disc cells that expressed genes associated with epithelized epiblast cells (Embryonic disc, n = 38). Further validation of cell lineages was also achieved by comparing our data to other human and non-human primate embryo datasets[28,64–66] (Figures 6C–6E and S5A–S4D).

(J) 12 d.p.f implantation site (left) and zoom view (right) of EVT cells (HLA-G+) migrating into scaffold away from the implantation site (2/2 embryos).
(K) Multinucleated STB cells (hCG+).
(L) 12 d.p.f implantation site and CTB cell columns (TP63+ and E-cadherin+) (2/2 embryos).
Images labeled (z-proj) display the max projection of z stack, and z-section represents a single plane. Scale bars: 20 μm (expanded panel in B), 50 μm (A, expanded panel in C), 100 μm (B, C, F, G, I, expanded panels in J, K, and L), and 200 μm (J, K, and L).
See also Figure S4 and Video S1.

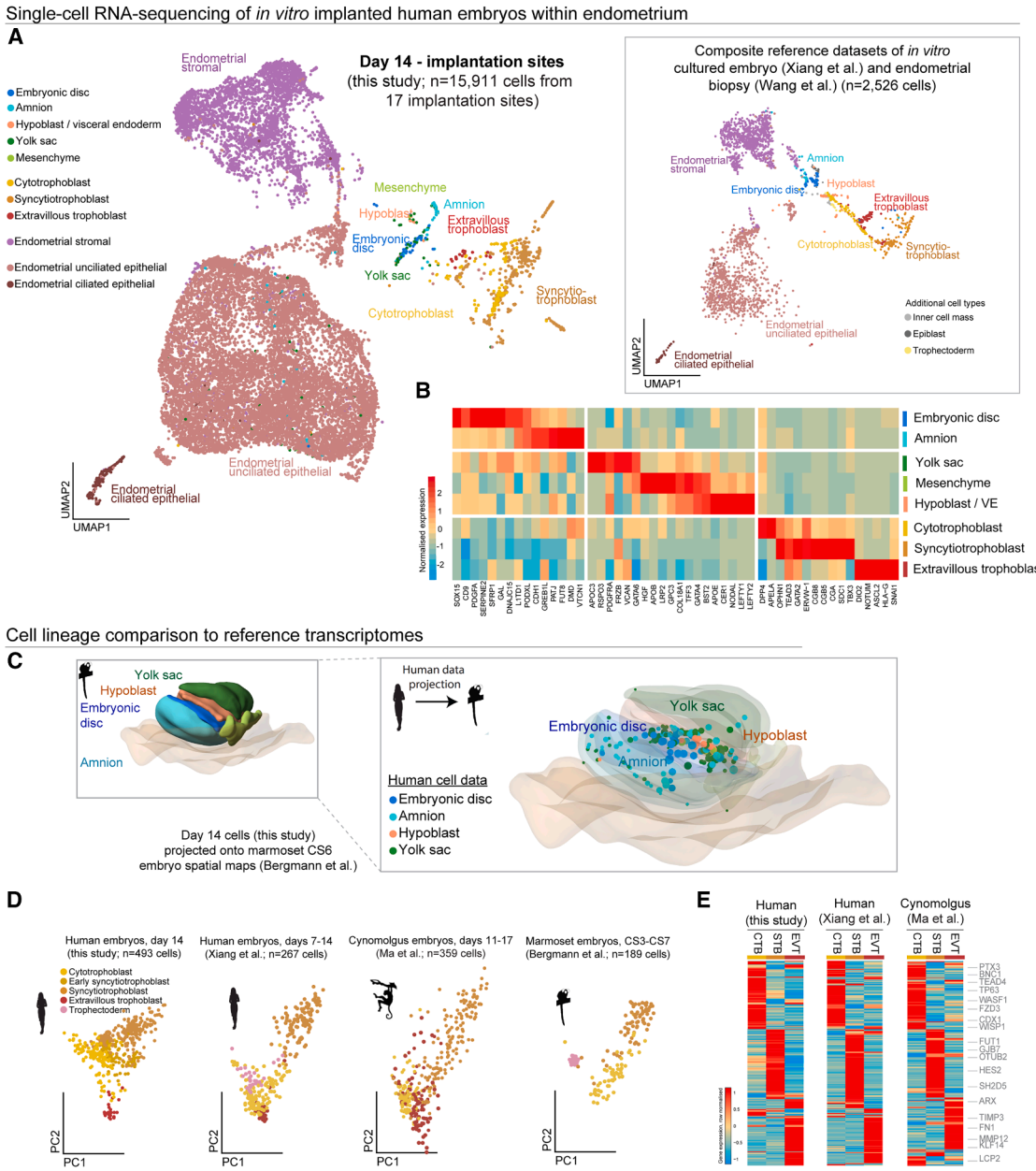

Figure 6. Single-cell transcriptional profiling of embryo implantation niche

(A) Uniform Manifold Approximation and Projection (UMAP) of scRNA-seq of 14 d.p.f. *in vitro* embryo implantation sites following alignment with reference datasets. Right, matched UMAP of composite reference datasets of *in vitro*-attached embryos[28] and endometrial biopsies.[47]

(B) Row-scaled (*Z* score) gene expression in our 14 d.p.f. dataset.

(C) Weighted nearest neighbor-based projection of scRNA-seq data from our study (represented by dots) onto a spatial transcriptome map of marmoset embryos.[64] Larger dot sizes indicate greater summed shared nearest neighbor (SNN) scores.

(D) Principal-component analysis (PCA) of trophoblast cells in our study (left) aligned to reference datasets from 7–14 d.p.f. *in vitro*-cultured human embryos,[28] 11–17 d.p.f. *in vitro*-cultured cynomolgus embryos,[65] and 15–25 d.p.f. marmoset embryos.[64]

(E) Genes differentially expressed between CTB, STB, and EVT in our study show similar patterns in trophoblast datasets from cynomolgus and human embryos.[65,66]

See also Figures S5 and S6.

Of the endometrial cells, our dataset contained stromal cells (*n* = 3,739) and epithelial cells (*n* = 11,215; Figures 6A and S5G), which were sub-classified into proliferative progenitors, secretory-luminal, ciliated, and secretory-glandular (Figures S5H–S5M and S6A–S6C). We used reference data from *in vivo* endometrium[67,68] to define gene signatures of stromal cells in

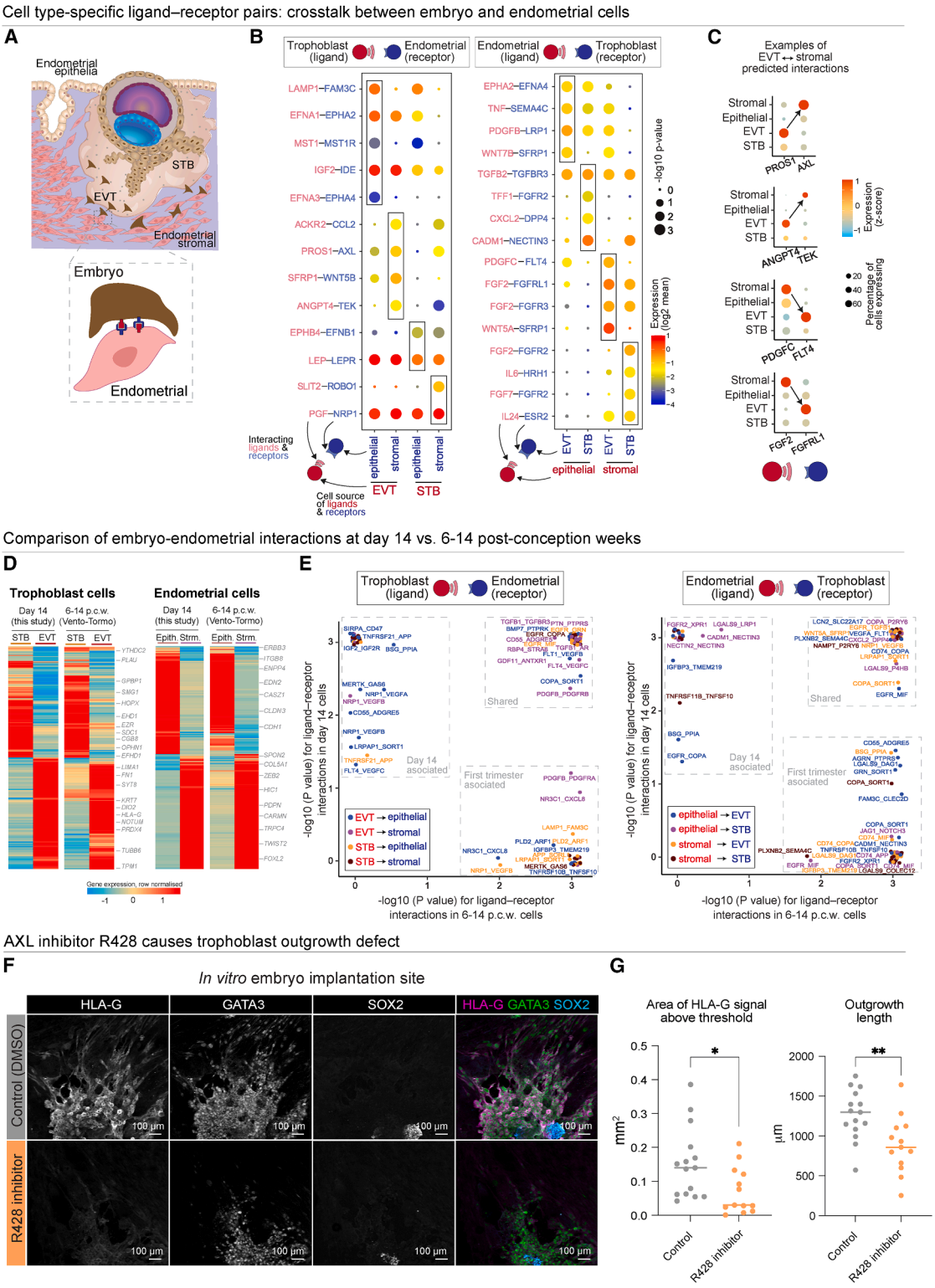

Figure 7. Identifying embryo-endometrial interactions within the implantation niche

(A) Schematic of implanting embryo.

(B) Ligand-receptor interactions between trophoblast (EVT or STB) and endometrial cells (epithelial or stromal). Dot plots show normalized, log-transformed average gene expression of selected ligands and receptors in indicated cell types. CellPhoneDB-reported *p* values in interacting cells are shown by dot sizes.

(legend continued on next page)

proliferative or decidualized states and used those signatures to annotate stromal cells in our *in vitro* implantation model. Most stromal cells in our samples corresponded to an *in vivo* decidualized state (Figures S6D–S6G). Together, these results resolve cell types within the *in vitro* embryo implantation niche at 14 d.p.f. and define their transcriptional profiles.

Dissecting embryo-endometrial crosstalk at implantation

Placentation and initiation of pregnancy are based on establishing communication between the embryo and surrounding uterine cells (Figure 7A).[54] To study the molecular communication initiated during implantation, we used our transcriptional dataset to predict ligand-receptor interactions between cell types within the embryo-endometrial interface. CellPhoneDB[67] analysis of trophoblast and endometrial cells revealed many cell-specific ligand-receptor pairs (Figures 7B and 7C), including an enrichment in tyrosine kinases and their ligands and regulators of migration and differentiation (Figures 7B and 7C).

To determine whether expression of identified ligands and receptors depends on co-culture of embryos with endometrium, we examined datasets of human embryos cultured without endometrial cells[28] and of *in vivo* endometrial biopsies.[68] Most ligand-receptor pairs showed similar patterns of interactions as in our embryo implantation dataset (Figure S7A). Interestingly, a small number of differences in ligand-receptor expression between our data and the reference data could be factors where tissue environment influences their induction (Figure S7A).

We then compared our dataset of the day 14 embryo implantation site with that of the maternal-fetal interface at 6–14 post-conception weeks (p.c.w.).[67] Unexpectedly, many of the gene profiles that define trophoblast cell types later in pregnancy[67] were already expressed in a lineage-restricted pattern in our dataset (Figure 7D), indicating substantial trophoblast lineage identity is established pre-gastrulation. We identified many (n = 84) shared ligand-receptor interactions across both datasets (Figure 7E, shared). These interactions have roles in decidualization and maternal-fetal tolerance (*TNFRSF10B/TNFSF10*, *CXCR6/CXCL16*, and *PTN/PTPRS*[69–71]) and in trophoblast development (*FLT4/VEGFC*, *FLT1/VEGFB*, and *IGF2/IGF1R*[72–75]). This suggests that many trophoblast-endometrial interactions are initiated early in development and persist throughout the first trimester of pregnancy. We also identified ligand-receptor interactions present at day 14 but not in the first trimester (n = 185) (Figure 7E, day 14 associated) as well as those present only in the first trimester (n = 73) (Figure 7E, first trimester associated). These findings uncover the order and dynamics in which embryo-endometrial interactions are established during early (14 d.p.f.) to late placentation (6–14 p.c.w.).

PROS1-AXL signaling between conceptus and endometrial cells is required for trophoblast outgrowth

To test interactions between embryo and endometrium, we focused on the identified *PROS1/AXL* interactions between EVT and stromal cells. AXL is a plasma membrane receptor tyrosine kinase, a member of the TAM family of kinases including TYRO3 and MERTK, and acts upstream of several signaling pathways, including phosphatidylinositol 3-kinase (PI3K)-AKT. We detected secreted PROS1 in culture media of implanted embryos (Figure S7B) and confirmed AXL expression in endometrial stromal cells (Figures S7C and S7D). Applying the inhibitor R428, which has higher selectivity for AXL compared with other TAM family members,[76] decreased phosphorylated AKT levels in endometrial stromal cells (Figure S7C). Importantly, treating implanted embryos with R428 caused a strong defect in the outgrowth of trophoblast cells within the stromal compartment (Figures 7F and 7G). Notably, inhibitor-treated cultures are reminiscent of the limited trophoblast growth of embryos developed in models lacking endometrial cells (Figure S7E). A defect in trophoblast outgrowth was also observed when implanted blastoids within the stromal compartment were treated with R428 (Figures S7F and S7G). By contrast, epiblast development appeared relatively unaffected (Figure S7H), and, importantly, the growth of separately cultured stromal cells and EVT was unaffected by the inhibitor (Figures S7I–S7L). Together, our results establish that inhibition with R428 disrupts trophoblast outgrowth during early embryo development *in vitro* and sheds light on the molecular crosstalk between conceptus and endometrium.

DISCUSSION

We provide an *in vitro* model of the receptive human endometrium that supports implantation and development of embryos, including the appearance of key structures in the embryo proper as well as advanced differentiation and organization of the trophoblast lineages. This model overcomes a long-standing barrier by enabling direct study of human embryogenesis and early placentation.

Assembled from primary cells isolated from endometrial biopsies, the model has several benefits, including (1) hormone-induced decidualization of the stromal compartment; (2) formation of luminal epithelium, which enables embryo apposition and adhesion; and (3) glandular invaginations that secrete histotrophic proteins to support embryo growth. We observed these

(C) Expression of selected ligands and receptors in endometrial and trophoblast cell types.
(D) Genes differentially expressed between EVT and STB (left) and between stromal and epithelial cells (right) in our study show similar patterns in corresponding cells from human embryo implantation sites at 6–14 p.c.w.[67]
(E) CellPhoneDB-reported p values for predicted ligand-receptor interactions identified in our study and in data from human embryo implantation sites at 6–14 p.c.w.[67]
(F) Images of *in vitro* embryo implantation site in control conditions (upper) and following treatment with AXL inhibitor R428 (lower). Representative of 2 independent experiments. Scale bars, 100 μm.
(G) Quantification of area of HLA-G signal above threshold (left) and quantification of embryo outgrowth length (right) in individual implantation sites (n = 15 implanted embryos for control; n = 13 implanted embryos for R428) from 2 independent experiments. p values from unpaired, two-tailed t test; $^*p < 0.05$.
See also Figure S7.

gland-like structures using live imaging, monitoring their formation of branched tubular networks. These changes demonstrate that our model recapitulates hormone-driven structural transformations observed in natural endometrium and provides a platform for investigating epithelial-stromal interactions, glandular morphogenesis, and potentially, functional secretions.

Our endometrial model permits implantation and post-implantation development of human embryos with organized lineages, and does so at a higher efficiency compared with other culture systems devoid of endometrial cells. Taking advantage of this, we provided insights into the timing, duration, and relative levels of protein secretions associated with the phases of embryo implantation. The increase of hPL between 12 and 14 d.p.f. suggests that secretion of this protein may be linked to establishing deeper trophoblast invasion. hPL plays a role in maternal metabolic adaptation to pregnancy, and its late-stage surge may reflect the increasing metabolic demands of the developing embryo. The increase in PAPP-A and PlGF at 10 d.p.f., followed by a sustained rise until 14 d.p.f., suggests an earlier involvement of these factors in trophoblast-endometrial interactions and may contribute to regulation of trophoblast proliferation and invasion at implantation.

Embryos cultured in our endometrial model show hallmarks of *in utero* embryos at CS5c-CS6a,[10–14] including bilaminar disc formation, expanded amniotic and yolk sac cavities, and putative amniotic cells and extraembryonic mesoderm. Of particular importance is the development of the trophoblast, which differentiated extensively and with greater organization compared with embryo culture systems that lack endometrial cells.[23–28] Observing the formation of the cytotrophoblastic shell, trophoblast cell columns and trabeculae, and early primary villi during culture of human embryos is an important advance because these are key landmarks in early placental development. Their formation *in vitro* underscores the necessary role of endometrial cells for promoting trophoblast differentiation,[22] thereby unlocking opportunities to investigate how the placenta is first formed.

Our model also supports implantation of stem cell-based embryo models. Following implantation, blastoid development showed some variability, the source of which may have origins in the cellular composition of the blastoids before implantation or in their ability to fully engage the post-implantation developmental program. Nevertheless, the compatibility of combining blastoids with the *in vitro* endometrial model provides experimental opportunities that would be challenging in human embryos, including ease of gene targeting for mechanistic studies and reporter-based live-cell imaging.

An advance of our model is the establishment *in vitro* of the early embryo-endometrial interface. Through transcriptome profiling, we identified ligand-receptor pairs enriched to specific cell types within the embryo implantation niche. While most interactions identified were stimulatory and associated with pro-survival signals, a few were inhibitory, indicating potential competition or mutual regulation between the growing embryo and surrounding endometrial tissues. An example of negative regulation was the EVT-expressed ligand *IGF2*, a known autocrine mediator of EVT migration,[77] and its endometrium-expressed interacting partner *IDE* (insulin/IGF2-degrading enzyme). The *IGF2–IDE* pair may suggest mutual restriction in growth between the developing embryo and surrounding endometrial tissue. We also demonstrated the functional importance of these interactions through perturbation experiments. Trophoblast outgrowth of embryos and blastoids within the stromal compartment was strongly affected by the AXL inhibitor R428. These results imply crosstalk, where factors released by the invading trophoblast are required for stromal cell behavior to support embryo growth. Interestingly, AXL levels are reduced in the decidua of patients with severe preeclampsia[78]; therefore, further studies on the effects of disrupted AXL signaling on trophoblast-endometrial interactions using CREST could shed light on the origins of this condition.

Our model and findings provide opportunities to understand human embryo implantation and the causes behind infertility, such as defective decidualization.[79] Based on this work, it is now possible to create patient-specific endometrial models that can be tested with a patient's embryos or induced pluripotent stem cell (iPSC)-derived blastoids. This could be valuable for personalized treatment options and in providing a platform for identifying compounds that promote embryo implantation. Enhancing implantation success through such strategies could help overcome a major barrier to pregnancy, reduce early preclinical losses, and improve live birth rates in both natural conception and assisted reproductive technologies.

Limitations of the study

(1) Our model does not attempt to recreate the full complexity of the endometrium. Future efforts could incorporate additional cell types, including immune and vascular cells, to more fully capture the complexity of the endometrial environment. (2) Engineering a thicker stromal compartment might support further endometrial glandular development and facilitate the modeling of deeper embryo invasion. (3) Our model is not straightforwardly adaptable to high-throughput assays or screens. (4) Although treatment of implanted embryos and blastoids with R428 caused defective trophoblast outgrowth, defining the roles of receptor tyrosine kinases within the TAM family in early human development requires further exploration.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Peter Rugg-Gunn (peter.rugg-gunn@babraham.ac.uk).

Materials availability

Unique/stable reagents generated in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

- RNA-seq data generated in this study have been deposited at ArrayExpress under accession number E-MTAB-13629 and are publicly available as of the date of publication. Processed data are available at Zenodo (https://doi.org/10.5281/zenodo.17930156).[80]
- All original code has been deposited at Zenodo (https://doi.org/10.5281/zenodo.17955321)[81] and is publicly available as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization: M.A.M. and P.J.R.-G.; data curation: C.P., A. Pokhilko, R.A.; formal analysis: M.A.M., S.E., I.Z., C.P., N.H., A. Pokhilko, M.P., A. Palomar, M.S., A.Q., A.J., S.M.-A., and R.Y.-S.; funding acquisition: M.A.M., L.A., L.N.-T., R.B.L., S.J.C., K.K.N., F.D., and P.J.R.-G.; investigation, M.A.M., S.E., I.Z., N.H., M.P., M.S., A. Palomar, M.S., Y.W., A.Q., C.A., K.B., A.J., J.S., A.B., S.M.-A., R.Y.-S., and A. Mercader; methodology, M.A.M., S.E., I.Z., N.H., M.P., A.J., and P.J.R.-G.; project administration, M.A.M., S.E., A. Palomar, F.D., and P.J.R.-G.; resources, M.A.M., S.E., I.Z., A. Palomar, A.Q., K.H., S.H., M.V., S.C., S.S., P.S., L.W., I.S., A. Mania, R.G., L.L., I.S.-R., A. Mercader, P.A., A.H.B., G.J.B., T.C.-D., R.C.F., L.A., L.N.-T., R.B.L., F.D., and P.J.R.-G.; supervision, M.A.M., S.J.C., K.K.N., A.R.D., F.D., and P.J.R.-G.; validation, M.A.M., S.E., I.Z., N.H., M.P., and A. Palomar; visualization, M.A.M., S.E., I.Z., C.P., N.H., A. Pokhilko, M.P., A. Palomar, R.A., and P.J.R.-G.; writing – original draft, M.A.M. and P.J.R.-G.; writing – review and editing, all authors.

DECLARATION OF INTERESTS

Babraham Institute filed a patent application describing the endometrial scaffold (WO2024184624A1; PCT/GB2024/050559). M.A.M. and P.J.R.-G. are inventors on this patent.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse monoclonal anti- Ki67 (85D)	Cell Signaling Technology	Cat# 9449, RRID: AB_2797703
Rabbit monoclonal anti- alpha Tubulin antibody (Acetyl-K40)	Abcam	Cat# ab179484, RRID: AB_2890906
Mouse monoclonal anti-Vimentin (5G3F10)	Santa Cruz Biotechnology	Cat# sc-66002, RRID: AB_1131285
Rat monoclonal anti-E-Cadherin (ECCD-2)	ThermoFisher Scientific	Cat# 13-1900, RRID: AB_2533005
Rabbit recombinant anti-Cytokeratin 7 [SP52]	Abcam	Cat# ab183344, RRID: AB_2936915
Rabbit monoclonal anti-Glycodelin A antibody [EP870Y]	Abcam	Cat# ab53289, RRID: AB_882116
Mouse monoclonal anti-Oct3/4 antibody (C-10)	Santa Cruz Biotechnology	Cat# sc-5279, RRID: AB_628051
Goat polyclonal anti-SOX17	R&D	Cat# AF1924, RRID: AB_355060
Goat polyclonal anti-GATA3	R&D	Cat# AF2605, RRID: AB_2108571
Rabbit recombinant anti-GATA3 [EPR16651] - ChIP Grade	Abcam	Cat# ab199428, RRID: AB_2819013
anti-HLA G [EPR23298-97]	Abcam	Cat# ab283260, RRID: AB_2924400
Rabbit monoclonal anti- Syndecan [SP152]	Abcam	Cat# ab130405, RRID: AB_11155360
Mouse monoclonal anti-hCG beta antibody [5H4-E2]	Abcam	Cat# ab9582, RRID: AB_296507
Rat monoclonal anti-SOX2 (Btjce)	ThermoFisher Scientific	Cat# 14-9811-82, RRID: AB_11219471
Rabbit monoclonal anti-TP63	Cell Signaling Technology	Cat# 13109 RRID: AB_2637091
Rabbit monoclonal anti-AP2 gamma/TFAP2C antibody [EPR20331]	Abcam	Cat# ab218107, RRID: AB_2891087
Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 488	ThermoFisher Scientific	Cat# A-21206, RRID: AB_2535792
Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 555	ThermoFisher Scientific	Cat# A-31572, RRID: AB_162543
Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 594	ThermoFisher Scientific	Cat# A-11012, RRID: AB_2534079
Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 405	ThermoFisher Scientific	Cat# A48257, RRID: AB_2884884
Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 488	ThermoFisher Scientific	Cat# A32766, RRID: AB_2762823
Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 555	ThermoFisher Scientific	Cat# A-31570, RRID: AB_2536180
Donkey anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	ThermoFisher Scientific	Cat# A-31571, RRID: AB_162542
Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	ThermoFisher Scientific	Cat# A-21447, RRID: AB_2535864
Donkey anti-Rat IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 405	ThermoFisher Scientific	Cat# A48268, RRID: AB_2890549
Donkey anti-Rat IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 488	ThermoFisher Scientific	Cat# A48269, RRID: AB_2893137
Goat anti-Rat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	ThermoFisher Scientific	Cat# A-21247, RRID: AB_141778
Phalloidin Alexa Fluor 594 Labeling Probes	ThermoFisher Scientific	Cat# A12381
Phalloidin Alexa Fluor 488 Labeling Probes	ThermoFisher Scientific	Cat# A12379

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Rabbit monoclonal anti-AXL (C89E7)	Cell Signaling Technology	Cat# 8661, RRID: AB_11217435
Rabbit monoclonal anti-TYRO3 (D38C6)	Cell Signaling Technology	Cat# 5585, RRID: AB_10706782
Rabbit monoclonal anti-P-ERK1/2	Cell Signaling Technology	Cat# 4695, RRID: AB_390779
Rabbit monoclonal anti-ERK1/2	Cell Signaling Technology	Cat# 5736, RRID: AB_10694237
Rabbit monoclonal anti-P-S473 AKT	Cell Signaling Technology	Cat# 4060, RRID: AB_2315049
Rabbit monoclonal anti-AKT	Cell Signaling Technology	Cat# 9272, RRID: AB_329827
Mouse monoclonal anti-ACTIN (β-Actin)	Sigma-Aldrich	Cat# A5441, RRID: AB_476744
Goat anti-rabbit HRP secondary antibody	Bio-Rad	Cat# 1706515, RRID: AB_11125142
Goat anti-mouse HRP secondary antibody	Bio-Rad	Cat# 1706516, RRID: AB_11125547
Mouse monoclonal anti-Vimentin	Cell Signaling Technology	Cat# 3390, RRID: AB_2216128
Mouse anti-HLA-G	Susan Fisher, UCSF	Cat# 48H4, N/A
Rabbit polyclonal anti-Cleaved Caspase-3	Cell Signaling Technologies	Cat 9661, RRID: AB_2341188
Mouse monoclonal anti-AXL (clone 108724)	Bio-Techne	Cat# MAB154, RRID: AB_2062558
Biological samples		
Human embryos (5 d.p.f.)	N/A	N/A
Donor-derived human endometrial (epithelial and stromal) cells	N/A	N/A
Chemicals, peptides, and recombinant proteins		
Advanced DMEM/F12	Gibco	12634-010
Collagenase type I	Sigma-Aldrich	Cat# SCR103
Dispase II	Sigma-Aldrich	Cat# D4693
Collagenase type IV	Sigma-Aldrich	Cat# C9263
Fetal Calf Serum (FCS)	Biosera	Cat# FB-1001
Charcoal-stripped Fetal Bovine Serum (FBS)	Sigma-Aldrich	Cat# F6765
Geltrex	Gibco	Cat# A1413301
B27 minus Vitamin A	Gibco	Cat# 12587-010
N-2 Supplement (100X)	Gibco	Cat# 17502-048
N-Acetyl-L-cysteine (NAC)	Sigma	Cat# A9165
Nicotinamide	Sigma	Cat# N0636
Recombinant human Rspondin-1	Peprotech	Cat# 120-38
Recombinant human FGF-10	Peprotech	Cat# 100-26
Recombinant human Noggin	Peprotech	Cat# 120-10c
Recombinant human EGF	Peprotech	Cat# AF-100-15
Recombinant human HGF	Peprotech	Cat# 100-39
A 83-01 (ALK-4,-5,-7 inhibitor)	Tocris	Cat# 2939
TrueGel3D Hydrogel Kit	Sigma-Aldrich	Cat# TRUE7
Collagen III	Rockland immunochemicals	Cat# 009-001-105
Collagen I	Rockland immunochemicals	Cat# 009-001-103
Collagen VI	Abcam	Cat# ab7538
Fibronectin	Merck Millipore	Cat# FC010
Oestradiol (E2) – water soluble	Sigma-Aldrich	Cat# E4389
Progesterone (P4) – water soluble	Sigma-Aldrich	Cat# P7556
8-bromoadenosine 3′, 5′-cyclic monophosphate (cAMP)	Sigma-Aldrich	Cat# B7880
Human Serum Albumin	Vitrolife	Cat# 10064
Acidic Tyrode's solution	Origio	Cat# 10605000A
Hyaluronan	Abcam	Cat# ab143634
Human prolactin (PRL)	Peprotech	Cat# 100-07

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human chorionic gonadotropin (hCG)	Alpha Diagnostic Intl	Cat# RP-1484
Human placental lactogen (hPL)	R&D	Cat# 5757-PL/CF
Insulin growth factor (IGF)	Cell Guidance Systems	Cat# GFH34AF
Y-27632 (ROCK inhibitor)	Cell Guidance Systems	Cat# SM02-10
PD0325901	Cambridge Stem Cell Institute	N/A
Human LIF	Cambridge Stem Cell Institute	N/A
Go6983 (PKC inhibitor)	Tocris	Cat# 2285
XAV939 (Tankyrase1/2 inhibitor)	Merck	Cat# 575545
Bemcentinib R428 (AXL inhibitor)	Selleckchem	Cat# S2841
Recombinant human PROS1 protein	Bio-Techne	Cat# 9489-PS-100
Recombinant Laminin iMatrix511	AMSbio	Cat# MS.892 012
CHIR99021	Cambridge Stem Cell Institute	N/A
SB431542 (ALK-4,-5,-7 inhibitor)	Bio-Techne	Cat# 1614
Valproic acid sodium salt	Sigma-Aldrich	Cat# P4543
Fibronectin	Bio-Techne	Cat# 1918-FN-02M
Human NRG1-β1 Recombinant Protein	Cell Signaling Technology	Cat# 26941S
KnockOut™ Serum Replacement (KSR)	Gibco	Cat# 10828-028
Critical commercial assays		
LDH-Glo Cytotoxicity Assay	Promega	Cat# J2380
Human Prolactin ELISA Kit	R&D	Cat# DPRL00
Insulin Like Growth Factor Binding Protein 1 ELISA Kit	Clone Cloud Corp	Cat# SEA052Hu
Human PAEP ELISA Kit	Abcam	Cat# ab275904
Free beta-Human Chorionic Gonadotropin ELISA Kit	Origene	Cat# EA100983
Human placental lactogen ELISA Kit	MyBioSource	Cat# MBS164143
Human PAPP-A ELISA Kit	R&D	Cat# DPPA00
Human placental growth factor ELISA Kit	Abcam	Cat# ab100629
Human Protein S ELISA Kit	Abcam	Cat# ab309109
Deposited data		
Human implantation sites (14 d.p.f. embryos implanted on CREST)	This paper	https://www.ebi.ac.uk/biostudies/ArrayExpress/studies/E-MTAB-13629?query=E-MTAB-13629; E-MTAB-13629
Human Reference Genome GRCh38	Genome Reference Consortium	https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.26/
Human pre-gastrulation embryos	Xiang et al.[28]	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE136447; GSE136447
Human endometrium	Wang et al.[47]	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111976; GSE111976
Proliferative and secretory endometrium	Garcia-Alonso et al.[68]	https://www.ebi.ac.uk/biostudies/ArrayExpress/studies/E-MTAB-10287?query=E-MTAB-10287; E-MTAB-10287
Endometrial stromal populations from 1st trimester pregnancy	Vento-Tormo et al.[67]	https://www.ebi.ac.uk/gxa/sc/experiments/E-MTAB-6701/results/cell-plots; E-MTAB-6701
Cynomolgus in vitro cultured embryo	Ma et al.[65]	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE130114; GSE130114
Marmoset CS5-7 in vivo embryos	Bergmann et al.[64]	https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-9367; E-MTAB-9367
Experimental models: Cell lines		
Human trophoblast stem cells (hTSC)	Okae et al.[82]	BTS11
Human Naive Embryonic Stem Cells (HNES)	Guo et al.[58]	HNES1

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human Naive Embryonic Stem Cells (HNES) with GFP	This paper	HNES1-pCAG-GFP
Software and algorithms		
Custom code and cell annotations for scRNASeq data of implantation sites (14 d.p.f. human embryos in CREST)	This paper	https://github.com/Rugg-Gunn-Lab/CREST
Olympus cellSens Software	Olympus	RRID:SCR_014551
Olympus scanR Software	Olympus	N/A
Fiji 1.53q	Schindelin et al.[83]	http://fiji.sc; RRID:SCR_002285
Leica Application Suite X (LASX)	Leica microsystems	https://www.leica-microsystems.com/products/microscope-software/details/product/leica-las-x-ls/; RRID:SCR_013673
CellProfiler Image Analysis Software (v4.2.1)	Broad Institute	http://cellprofiler.org; RRID:SCR_007358
10x Genomics Cellranger DNA (v.7.0.0)	10X Genomics	https://support.10xgenomics.com/single-cell-dna/software/pipelines/latest/what-is-cell-ranger-dna; RRID:SCR_023221
Seurat (v5.2.0)	Satija Lab	https://satijalab.org/seurat/get_started.html; RRID:SCR_016341
Souporcell	Heaton et al.[84]	https://github.com/wheaton5/souporcell
MAST (1.32.0)	Bioconductor	https://bioconductor.org/packages/release/bioc/html/MAST.html; RRID:SCR_016340
CellPhoneDB (v5)	Troulé et al.[85]	https://github.com/ventolab/CellphoneDB; RRID:SCR_01178
Pheatmap (1.0.12)	CRAN	https://www.rdocumentation.org/packages/pheatmap/versions/1.0.13; RRID:SCR_016418
Ggplot2 (v3.5.1)	CRAN	https://cran.r-project.org/web/packages/ggplot2/index.html; RRID:SCR_014601
Bcftools v1.7	SAMTOOLS	http://htslib.org/; RRID:SCR_002105
Harmony v1.2.3	CRAN	https://github.com/immunogenomics/harmony; RRID:SCR_02220
SpatialModelling	MATLAB	https://github.com/Boroviak-Lab/SpatialModelling
GraphPad Prism (V8)	GraphPad	http://www.graphpad.com/; RRID:SCR_002798
Imaris	BitPlane	http://www.bitplane.com/imaris/imaris; RRID:SCR_007370
ZEISS ZEN Microscopy Software	Zeiss	https://www.zeiss.com/microscopy/en/products/software/zeiss-zen.html; RRID:SCR_013672
Other		
Phenol red-free DMEM	Sigma-Aldrich	Cat#D4947
Kitazato Thawing Kit	Kitazato	Cat#VT802-2 or VT602
Vit Kit-Thaw by Irvine Scientific	Fujifilm	Cat#90137-SO
Embryo Culture Medium SAGE 1step	Origio	Cat#67010060A
Transwell ThinCert™ Tissue Culture Inserts	Greiner Bio-One	Cat#662630
Ibidi 8-Well High Glass Bottom μ-Slides	Ibidi	Cat#80807
24-well AggreWell™400 plate	STEMCELL Technologies	Cat 34450

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Ethics statements

Human embryos

The goal of this research study was to build and test an *in vitro* model of receptive endometrium with the aim of advancing understanding of human embryo implantation and early post-implantation development. All experiments related to the culture of

human embryos were performed according to the relevant institutional, national and international guidelines and regulations, as detailed below. Experiments conducted at the Babraham Institute (UK) were performed in accordance with the Human Fertility and Embryology Authority (HFEA) under the research licence number RO209-1-a. The process of licence approval entailed independent peer review along with consideration by the HFEA Licence and Executive Committees. The research of this study complies with the HFEA Code of Practice (9th Edition, revised in October 2023), updated laws governing the storage of gametes and embryos that came into effect on 1 July 2022 (including Health and Care Act 2022 and HFEA Clinic Practical Guide on legal changes to storage limits and guidance) and has been subject to inspections by the HFEA since the licence was granted. Ethical approval was obtained from the Health Research Authority's West Midlands – Black Country Research Ethics Committee (reference number 21/WM/0194). Experiments conducted at the IVI RMA Global Research Alliance – IVI Foundation Valencia (Spain) were performed in accordance with the following national and local ethical approvals: embryos recruited from IVI RMA Global Research Alliance – IVIRMA Valencia (under national Ethics Committee approval by National Commission of Human Assisted Reproduction reference numbers 32544/72212, 3/2019 and 6/2023, corresponding to local Ethics Committee approval reference numbers 1812-FIVI-102-FD, 1906-BCN-064-IS and 2304-FIVI-054-IS, respectively). Experiments conducted at Stanford (US) were performed in accordance with California Department of Public Health (Guidelines for Human Stem Cell Research Pursuant to Health and Safety Code 125118, updated April 2018) and CIRM Regulations (Chapter 2. Scientific and Medical Accountability Standards, § 100030, Activities Not Eligible for CIRM Funding, updated 08/21/17)). Only supernumerary embryos have been used, recruited by the Stanford RENEW Biobank (IRB 10466). Experiments were performed under protocols SCRO 929 and IRB 72398 approved by the Stanford Human Subjects Research Compliance Office. All embryo cultures at Stanford were stopped by 12 days or before the appearance of the primitive streak (whichever is earlier). All experiments were conducted in a research building independently funded by the California Institute for Regenerative Medicine, and no federal funding was used for these experiments. The evaluation and approval of this research through a specialised scientific and ethics review process complies with the International Society for Stem Cell Research (ISSCR) Guidelines for Stem Cell Research and Clinical translation (2021). Specifically, that the research has a compelling scientific rationale that necessitates the use of embryos and that the minimum number of embryos to achieve the scientific objectives were used.

Embryo donors were recruited from patients at the following IVF clinics: Cambridge IVF (Cambridge, UK), Centre for Reproductive and Genetic Health (London, UK), King's Fertility London (London, UK), IVI RMA Global Research Alliance – IVIRMA Valencia (Spain), Stanford RENEW Biobank. Informed consent was obtained from all couples who donated their supernumerary embryos following IVF treatment. Before giving consent, donors were provided with full information about the research project and objectives, potential risks and benefits, that participation to donate their embryos is voluntary and they can withdraw from the study at any time, how their information will be used, and they had the opportunity to receive counselling before agreeing donation. Donors were fully informed that embryo cultures would be stopped by 14 days post-fertilisation in the UK and Spain or by 12 days post-fertilisation in the US, that subsequent biochemical and genetic studies would be performed on the samples, and that the results of these will be published in scientific journals. No financial inducements were offered for donation. Before the start of each experiment and thawing of an embryo, we verify that patients' consents continue to be valid and that consent has not been withdrawn.

Human embryonic stem cells and blastoids

Research involving human embryonic stem cells has been approved by the Steering Committee of the UK Stem Cell Bank (reference SCSC11-58). Research involving the generation and *in vitro* implantation of human blastoids has been approved by the Human Biology Research Ethics Committee of the University of Cambridge (application number HBREC.2022.23), and by the Steering Committee of the UK Stem Cell Bank (reference SCSC22-10). Experiments conducted at Stanford were performed under protocol SCRO 919 approved by Stanford Stem Cell Research Oversight Panel in accordance with California State Regulations, CIRM Regulations and Stanford's Policy on Human Embryonic Stem Cell Research. All experiments were conducted in a research building independently funded by the California Institute for Regenerative Medicine, and no federal funding was used for these experiments.The assessment and approval of our research involving the generation of stem cell-based embryo models (blastoids) through a specialised scientific and ethics review process complies with the ISSCR Guidelines for Stem Cell Research and Clinical translation (2021). In keeping with these guidelines, and with the UK Code of Practice for the Generation and Use of Human Stem Cell-Based Embryo Models, blastoids were cultured for the minimum duration necessary to achieve the scientific objectives of this study. To compare blastoid development with embryo development, all blastoid experiments were stopped at or before a developmental stage equivalent to embryos at 12 d.p.f. (in Stanford, USA) or 14 d.p.f. (Cambridge, UK). This corresponded to a maximum of six and nine days, respectively, of *in vitro* culture starting from the day the blastoid was transferred into the endometrial model.

Endometrial samples

Collection of endometrial tissue biopsies for the derivation of epithelial and stromal cells used in this study was performed in the UK, Spain and the US according to the relevant institutional and national guidelines and regulations, as detailed below. In the UK, the donation of human endometrial tissues has been approved by the NHS Health Research Authority London – Hampstead Research Ethics Committee (reference 20/LO/0115). In Spain, the donation of human endometrial tissues has been approved by Comité de Ética de la Investigación IVI Valencia (under the project reference 2304-FIVI-054-IS) and agreed to be transferred and used at the Babraham Institute (research agreement BI21-922). In the US (Stanford), the donation of human endometrial tissues has been approved under IRB Protocol 67323 and 69554, by the Stanford Human Subjects Research Compliance Office.

Endometrial biopsies were collected from healthy women (n=19 donors). Donors were fertile women aged 18-31 years (UK, Spain) or 18-36 years old (US) with a body mass index (BMI) of 19-25kg/m^2, reporting normal menstrual cycles (21-35 days) and proven fertility. Samples were processed to isolate both endometrial stromal and epithelial fractions on the same day of biopsy collection.

Endometrial scaffolds and hormone stimulation

Isolation of cells from endometrial biopsies

Step-by-step protocols for CREST are available from protocols.io (https://dx.doi.org/10.17504/protocols.io.rm7vzx67xgx1/v1). Out of 17 endometrial biopsies collected from healthy donors, eight stromal and 13 epithelial donor-derived cell cultures were derived from endometrial biopsies. Isolation of endometrial cells (stromal and epithelial fractions) was performed by two methods: either by gravity sedimentation or by enzymatic digestion. For the isolation method by gravity sedimentation, the endometrial tissue was mechanically fragmented with scalpels. Fragments were enzymatically digested with 1 mg/ml collagenase type I (Sigma-Aldrich, SCR103) in phenol red-free DMEM (Sigma-Aldrich, D4947) and incubated overnight at 4°C. After incubation, samples were vertically oriented for 10 minutes, allowing separation of epithelial and stromal compartments according to different fragments size by gravity sedimentation. For the isolation method by enzymatic digestion, the endometrial biopsy was mechanically fragmented and enzymatically digested with a mixture of dispase II (1.25 U/mL, Sigma-Aldrich, D4693) and collagenase type V (0.4 mg/mL, Sigma-Aldrich, C9263) diluted in RPMI 1640 medium (Gibco, 21875-034) supplemented with 10% fetal calf serum (FCS, Biosera, FB-1001) and incubated for 30 minutes at 37°C with shaking. Supernatant was filtered (100 μm cell filters, Sigma-Aldrich, 431752) to retain the epithelial glandular fraction. Once the stromal and epithelial fractions have been isolated by either of the two methods, epithelial fragments were pelleted by centrifugation (600 x g, 6 minutes), resuspended in ice-cold Geltrex (Gibco, A1413301) and cultured in organoid medium as detailed below. Stromal cells were recovered from the supernatant after filtering (50 μm cell filters, Celltrics GmbH, Germany), pelleted after centrifugation (1000 x g, 5 minutes) and cultured in stromal medium as detailed below.

Culture of human endometrial stromal cells

After isolation, endometrial stromal cells were resuspended in Advanced DMEM/F12 medium (Gibco, 12634010). Homogeneity of cultures was determined according to morphological characteristics and verified by immunocytochemical localisation of the stromal marker Vimentin and the immune cell marker CD45. Stromal cells were seeded and propagated in 25 cm^2 or 75 cm^2 cell culture flasks (Nunc EasYFlask Cell Culture Flasks, Thermo Scientific, 156367) in the following medium referred as the "stromal medium (10% FBS)": Advanced DMEM/F12 medium supplemented with 10% of charcoal-stripped FBS (Sigma-Aldrich, F6765), 2mM L-glutamine (Gibco, 25030024) and 0.6 % antibiotics (Penicillin-Streptomycin 10000 U/mL-10000 μg/mL; Gibco, 15140122). Culture conditions were 37°C, 20% atmospheric O_2 and 5% CO_2. Medium was changed every 48h and when confluence reached 80% - 90%, stromal cells were passaged using Accutase (7 minutes at 37°C) and further propagated at ratio 1:10. Stromal cells between passages 2 and 8 were used to create endometrial scaffolds. Cells were routinely verified to be mycoplasma-negative.

Culture of human endometrial epithelial cells

After isolation, endometrial epithelial fragments were resuspended in Geltrex, seeded in 25 μL domes in a 48-well plate (Corning Costar, 3548) covered with 250 μL "organoid medium" modified from[40] and containing: Advanced DMEM/F12, (Gibco, 12634-010) supplemented with 1 x B27 minus Vitamin A (Gibco, 12587-010), 1 x N2 (Gibco, 17502-048), 2 mM L-glutamine (Gibco, 25030024), 0.2 % antibiotics (Penicillin-Streptomycin 10000 U/mL, Gibco, 15140122), 1.25 mM N-Acetyl-L-cysteine (NAC, Sigma, A9165), 10 mM Nicotinamide (Sigma, N0636), recombinant human Rspondin-1 (80 ng/mL, Peprotech, 120-38), recombinant human Fibroblast Growth Factor 10 (FGF-10 100 ng/mL, Peprotech, 100-26), recombinant human Noggin (100 ng/mL, Peprotech, 120-10c), recombinant human Epidermal Growth Factor (EGF, 50 ng/mL, Peprotech, AF-100-15), recombinant human Hepatocyte Growth Factor (HGF, 50 ng/mL, Peprotech, 100-39), ALK-4,-5,-7 inhibitor (A83-01, 500 nM, Tocris, 2939). Culture conditions were 37°C, 20% atmospheric O_2 and 5% CO_2. Medium was replaced every 48 hours and domes passaged at 1:3 ratio usually after 4–5 days depending on confluence. Passaging was performed by manual disaggregation as described previously[40]: domes were disaggregated and centrifuged (600 x g, 6 minutes); pellet was dissociated by manual pipetting (400 times) and suspension was centrifuged (600 x g, 6 minutes); pellet is then resuspended in fresh Advanced DMEM/F12 and dissociated (200 times) to obtain small fragments. After centrifugation (600 x g, 6 minutes), pellet containing endometrial epithelial fragments was resuspended in ice-cold Geltrex (25 μL per dome) and seeded in a 48-well plate, covered with 250 μL organoid medium. Cells were routinely verified to be mycoplasma-negative. Endometrial organoids were used between passages 2 and 20.

Assembly of endometrial scaffolds

Construction of CREST is a multi-step process and begins with the assembly of a stromal compartment inside a transwell (Greiner Bio-One ThinCert™ Tissue Culture Inserts, 662630), with two media reservoirs, one in the lower compartment and one in the upper compartment. We used a hydrogel backbone (TrueGel3D Hydrogel Kit, Sigma-Aldrich, TRUE7) containing dextran polymers and cyclodextrin crosslinkers. When first establishing the scaffold, we varied the ratio of dextran polymers and cyclodextrin crosslinkers to identify optimal conditions. In these experiments, cell stress was assessed by quantification of secreted lactate dehydrogenase (LDH-Glo Cytotoxicity Assay, Promega, J2380,) using a GloMax® Discover Microplate Reader (Promega). In parallel, the percentage of stromal cells with an elongated morphology was measured using microscopy. Based on these experiments, a ratio of 2:3 (dextran polymers to cyclodextrin crosslinkers) was used in subsequent experiments. During these initial experiments, we also found that it was beneficial to include ECM components to the stromal/hydrogel mix. To mimic the ECM composition of the natural endometrium,

the hydrogel was supplemented with Collagen III (90 μg/ml) (Rockland immunochemicals, 009-001-105), Collagen I (70 μg/ml) (Rockland immunochemicals, 009-001-103), Collagen VI (20 mg/ml) (Abcam, ab7538) and Fibronectin (20 mg/ml) (Merck Millipore, FC010). The dextran polymers, buffer solution and ECM molecules were incubated on ice for 1 hour to allow *in vitro* fibrillogenesis. After incubation, the pH of the solution was neutralised by adding 8.75 μl of stromal medium (10% FBS), HEPES (final concentration 25mM) (Gibco, 15630080) and 1M sodium bicarbonate (final concentration 76.67 mM). For each scaffold system, 100,000 stromal cells in 5 μL stromal media was premixed with 10% cyclodextrin (final) and then incorporated into the neutralised dextran extracellular matrix mix, a total volume of 30 μL. A drop of 30 μL containing the stromal cells was seeded at the centre of each transwell and allowed to fully polymerise for 25 minutes at 37°C to avoid formation of the meniscus. Once polymerised, 1 mL of stromal medium (10% FBS) was placed in the lower compartment of the transwell and 250 μL of stromal medium (10% FBS) in the upper compartment. Transwells were incubated at 37°C, 20% atmospheric O_2 and 5% CO_2 and media changed every 48 hours, until stromal cells reach full elongation (4-7 days). After stromal cells exhibited elongated fibroblast-like morphology, the endometrial epithelial cells are added to the system to induce formation of the endometrial luminal and glandular components. Endometrial epithelial organoids were passaged and dissociated as described above. Endometrial epithelial fragments (one initial dome, which contains ∼1,500 fragments, was seeded into two scaffolds) were resuspended in organoid medium supplemented with 2% charcoal-stripped FBS and seeded in the upper transwell compartment. The lower compartment was maintained in stromal media (10% FBS) due to the different nutritional requirements of the stromal and epithelial cells. Media in both compartments were replaced every 48 hours. Epithelial fragments give rise to outgrowths and were cultured until the cells formed a continuous epithelial monolayer covering the entire upper surface of the transwell (∼5-7 days). Assembled scaffolds were hormone stimulated within one week of epithelial outgrowth.

Hormonal stimulation of endometrial scaffolds

The scaffolds were stimulated hormonally in two steps to mimic the proliferative phase and mid-secretory phases of the human endometrial cycle. In the first step, the scaffolds were stimulated with 10 nM β-oestradiol (E2) (Sigma-Aldrich, E4389) for 4 days as follows: 250 μL of organoid medium (+ 2% FBS, + 10 nM E2) in the upper compartment; 1 mL of stromal medium (2% FBS, + 10 nM E2) in the lower compartment. Media were replaced after 48 hours. In the second step, scaffolds were stimulated with full hormonal stimulation for 8 days with 10 nM β-oestradiol (E2), 1 μM progesterone (P4) (Sigma-Aldrich, P7556) and 1 μM 8-bromoadenosine 3′, 5′-cyclic monophosphate (cAMP) (Sigma-Aldrich, B7880) as follows: 250 μL of organoid medium (+ 2% FBS, + 10 nM E2, + 1 μM P4 and 1 μM cAMP) in the upper compartment, 1 mL of stromal medium (2% FBS, + 10 nM E2, + 1 μM P4 and 1 μM cAMP) in the lower compartment. Media were replaced every 48 hours.

METHOD DETAILS

Characterisation of endometrial scaffolds

Stiffness measurement of the endometrial scaffold

Mechanical measurements of tissue stiffness were performed using a Bruker BioAFM. The *in vitro* endometrial scaffold was detached from the transwell system using sterile forceps and a scalpel. Care was taken to avoid damaging the tissue or the membrane during this process. The membrane was maintained at 37°C in 10% stromal medium in a sterile petri dish and immobilised by gluing cover slips at three edges on the periphery of the membrane. AFM cantilevers (MLCT-Spherical) with pre-calibrated nominal stiffness and a tip radius of 10 μm were used for recording force-displacement curves. Two cantilevers were used during these experiments. One had a stiffness of ∼0.087 N/m and resonant frequency of 8.39 Hz, while the other had a stiffness of ∼0.121 N/m, and frequency of 9.83 Hz. The loading and unloading rates were kept constant for all measurements at 2 μm/sec, and with maximum indentation depths of 1-3 μm. AFM cantilevers were cleaned in between measurements in an ethanol bath to remove any debris from previous tests. Young's moduli were estimated by fitting the approach part of the force-displacement curves with the Hertz model. In cases in which the adhesion force between the AFM tip and the sample was more than 5% of the maximum force, the retraction part of the force-displacement curves was fit to the JKR (Johnson–Kendall–Roberts) model, which accounts for the adhesion between the AFM probe and the tissue. The Poisson ratio of the samples was taken as 0.5. No correction for the effect of the substrate was made, as the indentation depth was less than 10% of the sample thickness for all cases. Raw data were processed with NanoScope Analysis to extract the Young's modulus values.

In vitro implantation of human embryos

Human embryo thawing

Supernumerary human blastocysts cryopreserved at 5 d.p.f. were thawed using one of two kits, following the manufacturer's instructions: either the Kitazato Thawing Kit (Kitazato, VT802-2 or VT602) or the Vit Kit-Thaw by Irvine Scientific (Fujifilm, 90137-SO). Blastocysts were handled using a micropipetter (Origio, MXL3-STR-CGR) and micropipetter flexible tips (Origio, MXL3-275). Open-pulled straws containing vitrified blastocysts were transferred directly from liquid nitrogen into 1 mL pre-warmed thawing solution (TS). After 1 minute immersed in TS, blastocysts were completely released from the open-pulled straw and transferred to a ReproPlate to continue thawing process (ReproPlates plate, Kitazato, Hunter Scientific, 83020). Blastocysts were transferred at room temperature consecutively into the following solutions Dilution Solution (DS, 3 minutes for Kitazato or 4 minutes for Irvine), Washing Step 1 Solution (WS1, 5 minutes for Kitazato or 4 minutes for Irvine), Washing Step 2 Solution (WS2, 1 minute for Kitazato or 4 minutes for Irvine). After these steps, the embryo is transferred in a single drop of Embryo Culture Medium SAGE 1step (Origio, 67010060A) supplemented

with 5% Human Serum Albumin (Vitrolife, 10064) covered by oil (Life Global, LGOL-100) and incubated for a minimum of 2 hours (ideally 5 hours) at 37°C, 20% O_2 and 5% CO_2 to allow time for spindle re-formation and recovery after thawing prior to subsequent manipulations. In this research study, we used a total of 248 human embryos at blastocyst stage (5-6 d.p.f.). This was the minimum number of embryos required to robustly test and evidence the fidelity of the *in vitro* receptive endometrial model and to collect the molecular and cellular data on embryo implantation and early post-implantation development that are presented in this article.

Human embryo culture in endometrial scaffolds

After incubation, only the recovered blastocysts that fully expanded were taken forwards and transferred onto endometrial scaffolds. For some embryos, the zona pellucida was removed by briefly treating the embryos with acidic Tyrode's solution (Origio, 10605000A). Human embryo medium (detailed below) (300 μl) was added to the upper compartment and 1 mL of stromal medium (stromal + 2% charcoal-stripped FBS supplemented with 10 nM β-oestradiol (E2) and 1 μM progesterone (P4)) was added to the lower compartment of the transwell and pre-equilibrated in an incubator for 3 hours prior to the transfer of the embryo. We devised the composition of the upper medium and named as Human Embryo Medium (HEM). The basal composition of HEM shares most of the supplements and growth factors (N2, B27, nicotinamide N-acetylcysteine, glutamine, and epidermal growth factor) with medium used to grow epithelial organoids and also to support the outgrowth and stimulation of epithelial fragments that are seeded on top of the stromal hydrogel. This composition fulfils requirements for the growth of epithelial cells. HEM also incorporates E2 and P4 to support the maintenance of the secretory phenotype of the endometrium. It also contains low levels of hCG, PRL, and hPL to maintain a basal hormone level. Hyaluronan contained in HEM creates a slightly viscous environment that encourages embryo contact with endometrial cells. Finally, IGF-1 was incorporated to support epiblast growth.[86] HEM-1 was used for the culture of embryos from 5 d.p.f. to 9 d.p.f.. HEM-1 (300 μL per scaffold) contains the following: Advanced DMEM/F12 (Gibco, 12634-010) supplemented with 1 x B27 (Gibco, 12587-010), 1 x N2 supplement (Gibco, 17502-048), 2 mM L-glutamine, 0.2 % Penicillin-Streptomycin (10000 U/mL, Gibco, 15140122), 1.25 mM N-Acetyl-L-cysteine (NAC, Sigma, A9165), 10 mM Nicotinamide (Sigma, N0636), 2% charcoal stripped FBS (Sigma Aldrich, F6765), 5% Human Serum Albumin (Vitrolife, 10064), 0.5 mg/mL hyaluronan (Abcam, Ab143634), 10 nM β-oestradiol (E2), 1 μM progesterone (P4), 20 ng/mL Prolactin (Peprotech 100-07), 1 μg/mL human Chorionic Gonadotropin (hCG, RP-1484, Alpha Diagnostic Intl) and 20 ng/mL human Placental Lactogen (hPL, R&D 5757-PL/CF), 50 ng/mL 1 IGF-1 (Cell Guidance Systems GFH34AF) and 50 ng/mL epidermal growth factor (EGF, Peprotech, AF-100-15). We used a modified version of HEM-1, termed HEM-2, to culture embryos from 9 d.p.f. to 12 d.p.f. due to increasing nutrient requirements of embryos. HEM-2 (350 μL) contains all compounds present in HEM-1 with the following modifications: the concentration of charcoal-stripped FBS was increased from 2% to 5%; and the following components were added: 7.5 mM glucose (Sigma, G8644); 1X Insulin-Transferrin-Selenium-Ethanolamine (Gibco, 51500-056), essential (1X), non-essential amino acids (1X) (Gibco 11130-036 and Gibco 11140-035) and 10 μM of ROCK inhibitor (Y-27632). A modified version of the media, termed HEM-3, is introduced at 12 d.p.f. (400 μL) and contains all compounds present in HEM-2 with the following modifications: the concentration of charcoal-stripped FBS was increased from 5% to 10%; the concentration of Human Serum Albumin was increased from 5% to 10%. HEM-2 and HEM-3 require pH adjustments using 1M sodium bicarbonate. Half-volume of each HEM was replaced daily with fresh medium (medium can be made and stored at 4°C for 4 days). 1 mL of stromal medium (2% charcoal-stripped FBS, with 10 nM E2 and 1 μM P4) was replaced daily with fresh medium in the lower compartment. Culture conditions were 37°C, 20% O_2 and 5% CO_2.

Exclusion criteria

In this study we used surplus embryos donated from couples undergoing IVF treatment. As a consequence, some embryos degenerated upon thawing or during the culture before reaching the desired final stage. These embryos were usually unable to attach and showed clear signs of cell death and were therefore excluded from the analysis.

Embryo culture models lacking endometrial cells

For the 2D attachment model, Ibidi 8-Well High Glass Bottom μ-Slides (Ibidi, 80807) were coated with 300 μL per well of 1% Geltrex in PBS for at least 1 hour at 37°C. The Geltrex was aspirated and replaced with 300 μL of HEM-1 per well and left to equilibrate at 37°C with 5% CO_2 for at least 1 hour. Single human blastocysts were transferred into individual wells of the μ-Slide and were cultured in HEM-1 from 5 d.p.f. to 9 d.p.f.. Media was then carefully replaced with HEM-2 and embryos were further cultured from 9 d.p.f. to 12 d.p.f., and then HEM-3 until 14 d.p.f. (in the UK and Spain only). For experiments with the '3D' culture system without endometrial cells, we created CREST scaffolds as described in the section 'Assembly of endometrial scaffolds' and omitted the endometrial epithelial and stromal cells. Single human blastocysts were transferred into individual transwells and were cultured in the same media as described in the section 'Human embryo culture in endometrial scaffolds.'.

Generation and *in vitro* implantation of blastoids

Culture of naïve human pluripotent stem cells

For the generation of blastoids in this study, we used the blastocyst-derived human naïve embryonic stem cell line HNES1.[58] In some experiments, we used a fluorescently labelled HNES1 line expressing pCAG-GFP-ires-hygro. Stable transgenic HNES1 expressing GFP were generated by transfecting cells with pCAG-GFP-ires-hygro plasmid with PB transposase expression vector pBase. 250,000 cells were dissociated as single cells with TrypLE and were co-transfected with pCAG-GFP-ires-hygro (550 ng) and pBase plasmid (550 ng) using FuGENE 6 (Promega, E2691), following the protocol for reverse transfection. For one well of a 12-well plate, we used 3.9 μL of transfection reagent, 1.1 μg of plasmids DNA, and 250,000 cells in 1 mL of PXGL medium with 10 μM Y27632 (ROCKi, Cell Guidance Systems, SM02-10). The medium was changed after overnight incubation and 200 μg/mL Hygromycin B

(Formedium, 101009) was added after 48 h. Cells were then selected for at least two weeks. HNES1 cells were cultured in PXGL medium containing N2B27 supplemented with 1 μM PD0325901 (Cambridge Stem Cell Institute), 10 ng/mL human LIF (Cambridge Stem Cell Institute), 2 μM Go6983 (Tocris, 2285) and 2 μM XAV939 (Merck, 575545). Cells were propagated on top of mitotically inactivated mouse fibroblasts supplemented with Geltrex. Cells were cultured at 37C in 7% CO_2 and 5% O_2. Karyotype analysis (20 cells per line, carried out by Cell Guidance Systems) showed that the wild-type HNES1 cells had a composite karyotype with a sub-clonal mix of 46XY and 45XY-chr10, and the GFP-expressing HNES1 cells are predominantly 45XY-chr10. Cells were routinely verified to be mycoplasma-negative.

Blastoid generation

For the induction of blastoids we used the method described below as modified from reference [33]. HNES1 were dissociated into single cells using TrypLE (Gibco, 12612604021 04021). MEF were removed using a dish coated with 0.1% gelatin, incubating cells for 50 minutes to allow attachment of MEF to the bottom of the dish. The supernatant with HNES1 cells was then resuspended in N2B27 medium supplemented with 1.5 μM PD0325901 (Cambridge Stem Cell Institute), 1 μM A83-01 (Tocris, 2939) and 10 μM Y-27632 (Cell Guidance Systems, SM02-10) and distributed so that 100,000 viable cells were seeded into each well of a 24-well AggreWell plate (AggreWell™400, STEMCELL Technologies). Each well contains a final volume of 2 mL. After 48 hours (day 2), 1.5 mL of the medium was replaced with fresh N2B27 containing 0.5 μM A83-01. After a total of 72 hours, the medium was replaced with N2B27 only. At 96 hours (day 4), the blastoid is fully formed. These structures approximate human embryos at 5 d.p.f. of development. In this research study, we used a total of 297 human blastoids.

Blastoid culture in endometrial scaffolds

Blastoids were transferred into the upper compartment of the endometrial scaffolds using a micropipetter (Origio, MXL3-STR-CGR) and micropipetter flexible tips (Origio, MXL3-275). Between 3 and 10 blastoids were transferred into each scaffold. Upper and lower compartment media of the transwell were pre-equilibrated in the incubator at 37C for 3 hours in the incubator before transfer of blastoids. Culture conditions were 37°C with 20% O_2 and 5% CO_2. The upper compartment contained 300 μL of HEM-1 medium; the lower compartment contained 1 mL of stromal medium (2% charcoal stripped FBS, with 10 nM E2 and 1 μM P4). Blastoids were cultured as human embryos above with the same media regime for up to 9 days. All blastoid experiments were stopped at or before a developmental stage equivalent to embryos at 12 d.p.f. (USA) or 14 d.p.f. (UK). This corresponded to a maximum of six and nine days, respectively, of *in vitro* culture starting from the day the blastoid was transferred into the endometrial model.

Embryo and blastoid implantation

Quantification of secretions

Quantification of secreted products was performed at different phases: before stimulation (unstimulated phase), proliferative phase after 4 days of E2 stimulation, in two different time-points of secretory phase (after 4 and 8 days of full hormonal stimulation with E2, P4 and cAMP) and during embryo culture between 5-14 d.p.f. Secretions were quantified by enzyme-linked immunosorbent assays (ELISA) separately for the top and bottom media corresponding, respectively, to the upper and lower compartments. The following secreted levels were quantified: PRL (Human Prolactin ELISA Kit, R&D Systems, DPRL00), IGFBP-1 (ELISA Kit for Insulin Like Growth Factor Binding Protein 1, Clone Cloud Corp, SEA052Hu), PAEP (Human PAEP ELISA Kit, Abcam, ab275904), b-hCG (ELISA kit for free beta-Human Chorionic Gonadotropin, Origene, EA100983), hPL (MyBioSource, MBS164143), PAPP-A (R&D Biosystems, DPPA00), PlGF (Abcam, ab100629) and PROS1 (Abcam, ab309109). Concentration of secreted proteins was calculated by mean value of replicates and normalising optical density measurements of each sample (either to organoid medium (2% FBS) for the upper medium or to stromal medium (2% FBS) for the lower medium, or to HEM media for the quantification during embryo culture), which were considered as blanks. All values shown for secreted proteins correspond to secreted levels by endometrial cells, human embryos or blastoids. Standard curves run for each protein tested were used to convert optical density measurements into concentration values.

Trophoblast stem cell culture

Maintenance of TSCs

Human trophoblast stem cells (TSCs, BTS11 cell line) were cultured as previously described[82] with minor modifications. A 6 well plate was coated with 0.5 μg/mL Recombinant Laminin iMatrix511 (AMSbio, AMS.892 012) in 1.5 mL of TSC medium [DMEM/F12 supplemented with 0.1 mM 2-mercaptoethanol, 0.2% FBS, 0.5% Penicillin-Streptomycin, 0.3%, BSA (Gibco 15260-037), 1% ITS-X (Gibco, 51500), 1.5 μg/mL L-ascorbic acid (Sigma A4544), 50 ng/ml EGF (Peprotech AF-100-15), 2 μM CHIR99021 (Cambridge Stem Cell Institute), 0.5 μM A83-01, 1 μM SB431542 (Bio-Techne, 1614), 0.8 mM VPA (Sigma, P4543), and 5 μM Y-27632] at 37°C for 30mins. Cells were seeded directly on the media supplemented with iMatrix for overnight. The day after seeding, the media was changed with fresh TSC media. Cells were cultured in 1.5 mL TSC medium in 5% CO_2 and 5% O_2. Media were changed every 2 days, and cells were passed using TrypLE Express every 3-4 days at a ratio of 1:8 or 1:10.

Differentiation of TSCs in EVTs

Differentiation of TSCs into terminal cell types was performed as previously described[82] with minor modifications. A 12-well plate was coated with 1 μg/mL Fibronectin (Bio-Techne, 1918-FN-02M) for at least one hour at 37°C. 50,000 TSCs were seeded per well in 1 mL EVT medium [DMEM/F12 supplemented with 0.1 mM β-mercaptoethanol, 0.5% penicillin-streptomycin, 0.3% BSA, 1% ITS-X, 2.5 μM Y-27632, 7.5 mM A83-01, 100 ng/mL NRG1 (Cell Signaling Technology, 26941S), 4% KSR (Gibco, 10828-028)]. Geltrex

was added to the final concentration of 2% after resuspending the cells in the medium. On day 3, medium was replaced with EVT medium without NRG1 and with Geltrex at 0.5% final concentration. To test toxicity of Bemcentinib R428 on EVTs, on day 5 of EVT differentiation, medium was replaced with EVT medium without NRG1 and KSR, with Matrigel at 0.5% final concentration, and with or without R428 for the following 2 days. The cells were ready for downstream analysis at day 7. Bright-field images were recorded using an EVOS M5000 microscope.

Quantitative PCR

Total RNA was isolated using the RNeasy Mini Kit (Qiagen, 74106) and complementary DNA (cDNA) was made from 500 ng of RNA using QuantiTect Reverse Transcription Kit (Qiagen, 205311). For quantitative PCR, SYBR® Green JumpStart *Taq* ReadyMix (Sigma-Aldrich, S4438) was used. Primers are: *NOTUM* (FOR: TGCTTCTTTGGCTACAAGGTC; REV: GGCACGTCCTTGAGTG TGT); *ITGA5* (FOR: GCCTGTGGAGTACAAGTCCTT; REV: AATTCGGGTGAAGTTATCTGTGG); *VGLL3* (FOR: GATTCCTGCTCCC CAGTGT; REV: TCAGTACCACGGTGATTCCTT). Three technical replicates were carried out. *GAPDH* was used as endogenous control to normalise gene expression. Quantitative PCR data were acquired with a Bio-Rad CFX384-5 machine and the Bio-Rad CFX Manager 3.1 software.

Imaging experiments

Live-cell imaging

For live cell imaging, fresh organoid medium supplemented with hormones and 2% charcoal-stripped FBS was prepared on the day. To 197 μL of this medium, 1 μL of SPY555 FastAct (1:10 dilution in DMSO) for a final dilution of 1:2,000 and 2 μL of SPY650 DNA (1:10 dilution in DMSO) for a final dilution of 1:1,000 were added. All medium from the top of the well was removed and replaced with the dye-containing medium. The scaffold was then incubated for 2 hours in normoxic conditions. After the 2-hour incubation, the organoid medium containing the dyes was removed and replaced with fresh organoid medium supplemented with 2% charcoal-stripped FBS and hormones. To maintain orientation during imaging, one foot of the transwell was marked with an ethanol-resistant marker pen. 5 mL of warmed Advanced DMEM/F12 was placed in the imaging dish, and the transwell was positioned in the centre. Samples were imaged using a confocal Zeiss LSM 700 microscope with the following objective: oil 20x/0.3 Plan Apo. Lasers (nm): 555 and 647. Images were acquired with the following parameters: resolution at 512 x 512 pixels, speed 800 Hz, Bi- direction X acquisition, Line average: 2, Pinhole 2.63AU, and sequential scanning between Z-stacks. A position map was created to track specific areas over time however, significant epithelial expansion after E2 stimulation made it challenging to locate the same positions. To address this, a 5x5 tile scan was performed. Low laser power was used to mitigate potential phototoxicity. Post-imaging, the scaffold was returned to the well, and the media were replaced. The top medium was completely replaced with organoid medium supplemented with 2% FBS and hormones, while the bottom medium was replaced with 2% stromal medium supplemented with hormones. Post-acquisition processing of raw image files was carried out using Fiji (version 2.9).

Bright-field imaging and staging of human embryos

Bright-field images of each embryo (top-down view using inverted microscope) during *in vitro* culture up to 14 d.p.f in the endometrial scaffolds were recorded using an inverted microscope (Axio Zeiss Vert A.1) coupled to a temperature-controlled plate. Embryo attachment was assessed and categorised during development by bright-field imaging. During stage 1 (hatching, 5 d.p.f), embryos are still hatching from zona pellucida and floating, which was ascertained by gently moving the tissue culture dish. In stage 2 (apposition, 6 d.p.f), the embryos have completed hatching and are making initial loose contact with the epithelial surface, orienting the inner cell mass towards the surface epithelium. Significant expansion of the blastocoel can be observed compared to stage 1. By stage 3 (adhesion, 7 d.p.f), the embryos are firmly attached to the luminal epithelial cells and show no movement with gentle agitation of the dish, but have not yet initiated implantation; for instance, the blastocoel cavity remains intact, and the structure is largely spherical. Finally, in stage 4 (implantation, ~8 d.p.f), implanted embryos are characterised by the proliferation and spreading of trophoblast cells into the stromal compartment, accompanied by the loss of the spherical features observed in earlier stages. Sankey plots were drawn using SankeyMATIC.[87]

Sample fixation

Samples were fixed in 4% paraformaldehyde (Electron Microscopy Sciences, 11586711) for 30 minutes at room temperature or overnight at 4°C. Samples were then washed in PBS containing 0.1% Tween (Sigma, P1379).

Preparation of cryosection samples

Comparative analysis shown in Figure 1I includes fixed tissues of one endometrial biopsy and CREST scaffolds (two biological replicates) hormonally stimulated to match native tissue. Each tissue was treated with 20% sucrose solution in PBS for 2 hours at 4°C. Tissues were then embedded in OCT (Fisher Scientific, 4585) and frozen using an isopentane solution cooled with dry ice. Tissue blocks were mounted on a Leica Cryostat (CM1860) and sectioned at 10-20 μm onto glass slides (Fisher Scientific, 1255015). Sections were then washed with warm PBS to remove OCT, fixed to the slide with 4% paraformaldehyde, and processed for immunofluorescence microscopy as described below.

Immunofluorescence microscopy

Samples were permeabilised in PBS containing 0.3% Triton X-100 and 0.1 M glycine for 30 minutes at room temperature. Samples were then blocked in blocking solution: PBS containing 10% FBS, 2% bovine serum albumin (BSA), and 0.1% Tween-20 for 1 hour at room temperature with rocking. Samples were then incubated overnight at 4°C in primary antibodies diluted in the above blocking solution (details below). The following day, samples were washed in PBS with 0.1% Tween-20 and incubated for 2 hours at room

temperature protected from light with fluorescently conjugated AlexaFluor secondary antibodies (details below) and DAPI (dilution 1/500, ThermoFisher Scientific, D3571), diluted in blocking solution. Samples were stored at 4°C in PBS supplemented with 0.1% sodium azide. We have not characterised or performed validation of primary antibodies but have chosen antibodies where target antigen, cross reactivity, validation by ELISA and Western Blot have been tested and provided by the manufacturer. Primary antibodies: acetylated alpha-Tubulin (Rabbit, 1:500, Abcam, ab179484), VIMENTIN (5G3F10, Mouse, 1:200, Santa Cruz Biotechnology, sc6602), VIMENTIN (Mouse, 1:200, Cell Signaling, 3390), E-CADHERIN (Rat, 1:200, ThermoFisher, 13-1900), CK7 (Rabbit, 1:200, Abcam, ab183344), PAEP/glycodelin (Rabbit, 1:500, Abcam, ab53289), OCT4 (mouse, 1:200, Santa Cruz Biotechnology, sc5279 or already conjugated sc5279 AF594, 1:100), SOX17 (Goat, 1:100, R&D, AF1924), GATA3 (Goat, 1:250, R&D, AF2605), GATA3 (Rabbit, 1:200, Abcam, ab199428), HLA-G (Rabbit, 1:100, Abcam, ab283260), HLA-G (Mouse, 1:200, 4H84, kindly provided by Susan Fisher at UCSF), SDC1 (Rabbit, 1:100, Abcam, ab130405), TP63 (Rabbit, 1:50, Cell Signaling 13109), hCG (Mouse, 1:100, Abcam, ab9582), SOX2 (Rat, 1:200, Invitrogen, 14-9811-82), TFAP2C (rabbit, 1:250, Abcam, ab218107) and Ki-67 (Mouse, Cell Signalling Technologies, #9449). Secondary antibodies were: Alexa Fluor 488 anti-Rabbit (1:500, ThermoFisher Scientific A21206), Alexa Fluor 555 anti-Rabbit (1:500, ThermoFisher Scientific A31572), Alexa Fluor 594 anti-Rabbit (1:500, ThermoFisher Scientific A-11012), Alexa Fluor anti-mouse 405 (1:500, ThermoFisher Scientific A48257), Alexa Fluor anti-mouse 488 (1:500, ThermoFisher Scientific A32766), Alexa Fluor anti-mouse 555 (1:500, ThermoFisher Scientific A-31570), Alexa Fluor 647 anti-Mouse (1:500, ThermoFisher Scientific A31571), Alexa Fluor 647 anti-Goat (1:500, ThermoFisher Scientific A21447), Alexa Fluor 405 anti-Rat (1:500, ThermoFisher Scientific, A48268) Alexa Fluor 488 anti-Rat (1:500, ThermoFisher Scientific A48269), Alexa Fluor 647 anti-Rat (1:500, ThermoFisher Scientific A-21247) together with Phalloidin Alexa Fluor 594 Phalloidin (1:250, ThermoFisher Scientific, A12381) or with Phalloidin Alexa Fluor 488 (1:250, ThermoFisher Scientific, A12379).

Image acquisition and processing

Samples imaged at the Babraham Institute Imaging Facility were imaged using a confocal Leica Stellaris 8 microscope with the following objectives: air 2.5x/0.07 Fluotar, air 10x/0.4 Plan Apo, air 20x/0.75 Plan Apo, oil 40x/1.4 Plan Apo, oil 63x 1.4 Plan Apo. Lasers (nm): 405, 488, WLL (440-790). Detectors: HyD-S, HyD-X, Transmitted light PMT. Images were acquired with the following parameters: resolution at 1024 x 1024 pixels, speed 400 or 600 Hz, Bidirectional X acquisition, Line average: 2, z-step size between 0.6 or 1.5 μm, Pinhole 1AU, sequential scanning between Z-stacks. Samples were also imaged using an Olympus SpinSR microscope with following objectives: air 4x/0.16, 10x/0.4. Lasers (nm): 405, 488, 561,640. For the images acquired in the Olympus SpinSR, the Olympus cellSens and Olympus ScanR software were used. Post-acquisition processing of raw image files was carried out using Fiji 1.53q software[83] or brightness adjustments, cropping, outlier removal and quantifications. LAS X software was used for 3D and volumes rendering. Samples imaged at Stanford University were imaged using a confocal Zeiss LSM700 confocal microscope with the 20x/0.8 Plan-Apo and oil 40x/1.3 Plan-Apo objectives. Lasers (nm): 405, 488, 555, 639. Detectors: two PMT, Transmitted light PMT. Images were acquired with the following parameters: resolution at 1024 x 1024 pixels, speed 800 Hz, Bidirectional X acquisition, Line average: 2, z-step size between 0.6 or 1.5 μm (or optimal selection), Pinhole 1.3AU, sequential scanning between Z-stacks. Post-acquisition processing of raw image files was carried out using Fiji version 2.9 software[83] or brightness adjustments, cropping, outlier removal and quantifications. Zen Black Edition and Imaris software packages were used for 3D and volumes rendering.

Image quantification

Morphometric analysis was performed on epithelial compartments at different phases: before stimulation (unstimulated phase), proliferative phase after 4 days of E2 stimulation and in two different time-points of secretory phase (after 4 and 8 days of full hormonal stimulation with E2, P4 and cAMP). CK7 was used as an epithelial marker. Cell segmentation was performed in the mid-section of a Z-stack, using Fiji 1.53q. Epithelial cells were segmented manually using polygon selection tool in Fiji: cell contour was drawn with multiple points to accurately fit perimeter of each cell segmented and Fit spline option was used to smooth cell contour and maximise the match of manual cell segmentation with cell perimeter. Each Region of Interest (ROI) corresponding to each epithelial cell segmented was added to ROI manager.

Quantifications of blastoid implantation, blastoid epiblast area, and embryo epiblast area were similarly performed, using the polygon selection tool to define the perimeter surrounding a region of GFP+ or OCT4+ cells for quantification of implantation and epiblast areas, respectively. The fit spline tool was used to smooth the curve, and measurements of area and height were collected.

For quantification of epiblast area for embryos from the Carnegie collection (histological sections), CS5b (embryo #8004), CS5c (embryo #7700) and CS6 (embryo #7801) embryos were selected for quantification. Sections through the epiblast were saved from the Virtual Human Embryo database (https://www.ehd.org/virtual-human-embryo/) and imported to Fiji as an image series. Manual alignment of the images was performed in Fiji if necessary. Scale was set using Fiji, based on a scale bar manually created for each specimen using the measurement tool in the Virtual Human Embryo database. A max projection was created through several planes in the epiblast center. Epiblast area was quantified as above.

Quantification of embryo outgrowth following R428 inhibitor treatment was performed in Fiji v1.54p (with Java 1.8.0_322). To generate a mask of the embryo outgrowth, an approximate outline was first drawn using the freehand selection tool to exclude fluorescence debris. This selection was made with reference to the maximum intensity z-projection of both the HLA-G and DAPI channels. The selected area was then segmented using Otsu intensity thresholding, with remaining objects filtered to exclude non-cellular particles smaller than 100 pixels. From the resulting mask, the smallest convex shape that encapsulated the embryo was calculated,

then the best-fitting ellipse for this convex hull was computed. The major axis length of this ellipse was taken as an estimate of the embryo length. The total area of HLA-G signal was measured using Analyze Particle tool.

Quantification of the epiblast area (SOX2+ cells) in the embryo following R428 inhibitor treatment was performed in Fiji v1.54p (with Java 1.8.0_322) using a semi-automated macro on maximum intensity projections of the 3D image stack. Due to complex fluorescence from debris outside of the feature of interest, a rectangular selection was drawn around the epiblast. Within this selection a Gaussian blur (radius = 1 pixel) was applied to smooth the image prior to auto-thresholding with the Otsu method. The resulting binary mask was used to measure the area.

Quantification of blastoid outgrowths following R428 inhibitor treatment was performed using Fiji 1.54f. Regions of interest (ROI) around the implantation sites were drawn using the polygon selection tool. HLA-G signal within the ROI was segmented according to a defined threshold value that was constant across all images. The total area of HLA-G signal above threshold was measured using Analyze Particle tool. The number of stromal cells was measured using Cell Profiler (v4.2.1).

Inhibition of receptor ligand interactions

Cell culture

To test the toxicity of R428 (Selleckchem, S2841), endometrial stromal cells were treated with a range of concentrations (500 nM - 2.5 mM) for 96 hours in culture (500 nM has previously been shown to affect migration of papillary thyroid carcinoma cells[88]) or the equivalent DMSO volume as a control. We observed no cell toxicity at these concentrations to endometrial stromal cells, and all further experiments used R428 at 500nM. Once embryos and blastoids had invaded into the stromal cell layer, 24 hours later 500nM R428 or equivalent DMSO was added fresh daily for 4 days into the appropriate HEM as standard culture for human embryos.

Western blotting

Endometrial stromal cells were treated for 15 minutes with recombinant human PROS1 protein (1μg/ml; Bio-Techne, 9489-PS-100) and/or R428 (500 nM). After treatment, cell culture medium was aspirated and dishes placed on ice. Cells were washed with ice-cold PBS and lysed in TG lysis buffer (20mM Tris-HCl pH7.5, 1mM EGTA, 137 mM NaCl, 1% v/v Triton, 10% v/v glycerol, 1.5mM MgCl2, 1mM Na3VO4 10μg/ml aprotinin, 10μg/ml leupeptin, 50mM NaF, 1mM PMSF). The collected lysates were frozen in liquid nitrogen and stored at -80°C. Lysates were thawed on ice then centrifuged at 13,000rpm for 10 minutes at 4°C. A Bradford assay was carried out to measure protein concentration. After boiling in Laemmli sample buffer, equal amounts of proteins were resolved by SDS-PAGE. Proteins were transferred to PVDF membrane (Millipore, IPVH00010). The membrane was blocked in 5% BSA/TBST for 1 hour, before overnight incubation with primary antibody at 4°C with gentle rocking. Blots were washed in TBST (3 x 5 minutes) before incubation with the appropriate secondary antibody diluted 1:3000 in 5% non-fat milk/TBST. Blots were washed in TBST (3 x 5 minutes) before detection with ECL (Cytiva, RPN2106). Antibodies were: AXL (rabbit, Cell Signaling Technology, 8661), TYRO3 (rabbit, Cell Signaling Technology, 5585), P ERK1/2(rabbit, Cell Signaling Technology, 4695), ERK1/2 (rabbit, Cell Signaling Technology, 5736), P S473 AKT (rabbit, Cell Signaling Technology, 4060), AKT (rabbit, Cell Signaling Technology, 9272), ACTIN (mouse, Sigma-Aldrich, 5441), goat anti-rabbit HRP (Bio-Rad, 1706515), goat anti-mouse HRP (Bio-Rad, 1706516).

Flow cytometry

Endometrial stromal cells were dissociated using Accutase and passed through a 50-μm cell strainer. Cells were washed with flow buffer (5% FBS in PBS) and incubated for 30 minutes at 4°C with AXL antibody (Bio-Techne, MAB154) using 250 ng antibody per million cells in 100 μL of flow buffer. Cells were washed with flow buffer and incubated with 0.5 μL donkey anti-mouse Alexa Fluor 488 secondary antibody (ThermoFisher Scientific, A-21202) per 500,000 cells in 200 μL volume for 30 minutes at 4°C in the dark. Cells were washed with flow buffer and analysed at the Babraham Institute Flow Core with a BD LSRFortessa cell analyser. Unstained cells were used in cytometer and gating setup and data were analysed using FlowJo V10.1 software (BD Biosciences).

Single-cell RNA-sequencing

Single cell isolation

Embryos were cultured from 5 d.p.f. as above until 14 d.p.f. At the end of the culture, each implantation site was dissected using precision forceps and scissors in warm DMEM. The implantation site was washed with PBS devoid of Mg^{2+} and Ca^{2+}. Enzymatic digestion was achieved using 50 μl of AccuMAX (Merck, SCR006) for 7 minutes at 37°C. AccuMAX contains collagenases which degrades the hydrogel releasing stromal and embryo cells. Single cell dissociation was achieved by manual pipetting using a micropipetter (Origio, MXL3-STR-CGR) and micropipetter flexible tips with decreasing apertures from 250 μm diameter (Origio, MXL3-275) to 50 μm diameter (MXL3-50, Origio). The reaction was stopped by transferring cells to ice cold PBS devoid of Mg^{2+} and Ca^{2+} and supplemented with 10% FBS. Following this, cells from each implantation site cells were labelled using the 10X Genomics Chromium 3' CellPlex Kit Set A (10x Genomics Chromium PN-1000261). Pooled single-cell suspension samples were loaded onto a Chromium Next GEM Chip G (10X Genomics® Chromium Next GEM Single Cell 3' Reagents Kits v3.1). Reverse transcription and library construction were carried out according to the manufacturer's instructions. Libraries were sequenced on Illumina NovaSeq 6000 S2 lane at the Cancer Research Cambridge Institute Genomics Facility. Samples were sequenced using a paired-end sequencing run with: read 1: 28 cycles; i7 index: 10 cycles, i5 index: 10 cycles; read 2: 90 cycles

Sequencing data processing and annotation

Sequenced data were aligned to human genome (GRCh38) with CellRanger v7.0.0 using a two-step process, setting 'force-cells' according to the elbow plot QC of the first run to select only high-quality cells. A composite reference dataset was created by merging

single cell RNA-sequencing data from[28,47] (GSE136447, GSE111976) for an initial alignment. This composite contains a ground truth reference for *in vitro*-cultured embryos and *in vivo* endometrial epithelial and stromal cells, as no existing combined reference exists for an early-stage implanted embryo with corresponding endometrium in human. Updated annotations for the embryonic cells of dataset[28] were taken from.[89] Our 10X data was jointly aligned to the reference dataset using Seurat[90] (v.5.2.1) IntegrateData function (with 4000 integration markers).

Louvain algorithm (Seurat FindClusters) was used on the integrated data for various resolutions (r=0.5,1,1.5,3) and later interpreted based on the consistency of the cell annotations (expression of cell-type markers) across the reference and our datasets, i.e. clusters annotated as trophoblast were enriched for trophoblast cells in the reference datasets, and clusters annotated as general trophoblast at a finer Louvain resolution represented a subset of those annotated as trophoblast at the lower resolution. The broad identity of the cells (embryonic, epithelial and stromal identity) was assessed using genetic differences between the cellular populations in each sample run, with help of Souporcell software.[84] 1-of-N genotypes (clusters) were set up according to the expected number of cell types in each sequenced batch (N=4 for samples 1,2,6,7,8 and N=6 for sample 3 and 3B). Note that at this stage the genotype label is arbitrary (e.g., cells will have a label {G1, …, G6}) and final cell type annotation (embryonic, stromal, and epithelial) was assigned by visualising the position of these genotypes in the aligned (expression) PCA spaces. After determining the broad tissue types, individual lineages were assigned based on their cluster identity and known markers. Independent batch correction of our dataset without the composite reference using either Seurat's IntegrateData or Harmony[91] showed a similar separation into three main components: embryonic, stromal, and epithelial, which were subsequently analysed separately.

Trophoblast cells were found to separate out from other embryonic lineages over all resolutions used in the Louvain clustering algorithm, and could be broadly annotated as trophoblast based solely on cluster ID. More refined annotations were assigned based on expression of a panel of markers (*KRT19, TACSTD2, KRT7, HLA−C, CDH1, CCKBR, TP63, CTNNAL1, VIM, ASCL2, JAM3, ELF5, OTX2, NOTUM, HLA−G, MKI67, FN1, LAMC1, DNMT3A, FSTL3, SPARC, ENPEP, FSTL1, GATA3, TEAD3, GATA2, SDC1, DPPA3, TFAP2C, TFAP2A, SNAI1, OVOL1, CGB5, CGA, CGB8*) and downstream alignment to other annotated datasets (see below). The panel of trophoblast markers were selected from known trophoblast markers from.[28,29,64] For non-trophoblast related embryo lineages, more refined annotation was assigned using a panel of markers for epiblast, amnion, visceral endoderm, secondary yolk sac, and extraembryonic mesoderm compiled from[28,64,92] (*APOA1, BAMBI, GATA6, GATA4, LGR5, APOB, LINC00261, GABRP, PRDM1, SOX15, CLDN4, CD9, FOXF1, EZR, SLC3A2, SOX17, VIM, HGF, RIMS2, BST2, COL6A1, COL6A2, COL6A3, COL1A1, NAV3, COL5A2, COL3A1, CGA, NLGN4X, TFAP2C, LINC01194, LUM, OPHN1, DCN, TNC, MT1G, MT1H, OTX2, LEFTY1, HHEX, HAND2, DPP4, PRLR, VTCN1, SNAI1, JARID2, CDH1, DPPA4, TCIM, COL1A2, WNT6, TANC2, TFAP2A, DNAJC15, CLDN6, SLC2A3, DPPA3, CLU, SFRP1, PODXL, PDGFA, GATA3, PDGFRA, LHX1, LEFTY2, POU5F1, CER1, SPARC, CST3, FN1, CASC15, DNMT3A*) and by comparison to other annotated datasets.

Heatmaps for populations of interest were generated using the AverageExpression function in Seurat, along with the pheatmap toolbox (v.1.0.12) or as individual datapoints using ggplot2. Heatmaps were plotted based upon the RNA gene expression slot and using row normalised data, unless otherwise indicated.

Cross-species alignment of cells

Final refinement of individual cell annotations was achieved by aligning to appropriate datasets. For embryonic and extraembryonic lineages, cells were jointly integrated with embryonic and extraembryonic cells from human,[28] cynomolgus,[65] and marmoset[64] using Seurat's IntegrateData (using CCA and 5000 anchor features). Once integrated, cells could be spatially projected onto a 3D CS6 marmoset embryo based on their proximity to other cells in the SNN graph using custom code and the SpatialModelling datasets.

Comparison of the trophoblast lineages of our dataset to those in human, cynomolgus, and marmoset was achieved by aligning the trophoblast-components of these annotated datasets using Seurat, with presumed hypoblast-derived lineages (hypoblast, yolk sac, extraembryonic mesoderm) similarly aligned to their corresponding cells.

Analysis of the stromal and epithelial cell populations was undertaken separately. Refined annotations for stromal samples were achieved by first visualising stromal lineages aligned to corresponding reference cells from proliferative and secretory endometrium[68] (E-MTAB-10287). Using the *in vivo* reference dataset, differentially expressed (DE) genes from proliferative vs decidualised stromal sub-populations were identified using FindMarkers (using MAST). Cells from our dataset were subsequently assigned a score based on the top 100 markers from this DE list using the AddModuleScore. Expression of additional markers were visualised using Seurat's VlnPlot or DotPlot alongside proliferative and secretory stage samples from the reference dataset[68] (E-MTAB-10287), as well as stromal populations from first trimester endometrium[67] (E-MTAB-6701).

For epithelial populations, putative ciliated cells formed distinct sub-clusters in the initial alignments to the composite reference (GSE136447/GSE111976), strongly expressed ciliated markers *PIFO*, and required no further re-annotation. Refined annotations for the remaining epithelial cells were achieved based on expression of key markers and comparison to annotated datasets. Preliminary alignment to cells from proliferative and secretory endometrium[46] (E-MTAB-10287) showed good overlap between our dataset and the main epithelial subpopulations found in cycling-endometrium (luminal, glandular, ciliated, and SOX9-enriched), with final lineage assignment based on cluster identity.

Downstream analysis of cell-cell interactions was performed on the cells that were further filtered based of the QC, expressing $\geq$ 2000 and $\leq$ 10000 genes/cell and having $\leq$ 30 % UMIs assigned to mitochondrial genes and $\leq$ 40 % assigned to genes coding ribosomal proteins. Within sequenced datasets the stromal populations were common to all scaffolds, whilst three separate endometrial epithelial populations were used, and each embryo should represent a unique genome. Individual genotypes in the VCF files were

integrated using bcftools and projected into 2D using PCA, which revealed a concentrated clustering of the stromal populations, but not of other populations, thus validating the genomic assignments.

Prediction of cell-cell interactions

To predict cellular interactions between embryo and endometrial cells in our dataset we used CellPhone database of ligand-receptor interactions,[67] under default parameters (https://github.com/Teichlab/cellphonedb). For this analysis we used expression matrixes of protein coding genes. The top specific pairs of ligand-receptor interactions were chosen based on their significance (CellPhoneDB interaction P value $\leq$ 0.05) and a specific expression in endometrial cells (stromal or epithelial) and embryo (EVT or STB cell types). First trimester data[67] were downloaded from https://www.reproductivecellatlas.org.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed using GraphPad Prism v8 software unless otherwise indicated. The definition of n for each experiment is provided in the figure legends and corresponds to independent biological replicates (for *in vitro* assays) or individual embryos/blastoids (for implantation studies). For comparisons involving more than two groups, data were analysed using the Kruskal–Wallis test followed by Dunn's post-hoc correction. Two-group comparisons were assessed using either a Mann–Whitney test (for non-parametric data) or an unpaired, two-tailed t-test (for parametric data). Categorical outcomes, such as blastoid or embryo implantation success and epiblast organisation, were analysed using Fisher's exact test. In morphometric analyses of epithelial cells, outliers were identified and removed using the ROUT method (Q = 5%). Data are represented as boxplots showing median, interquartile range, and minimum–maximum values, or as mean with individual data points, as specified in the figure legends. Statistical significance was defined as $p<0.05$.

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Supplemental figures

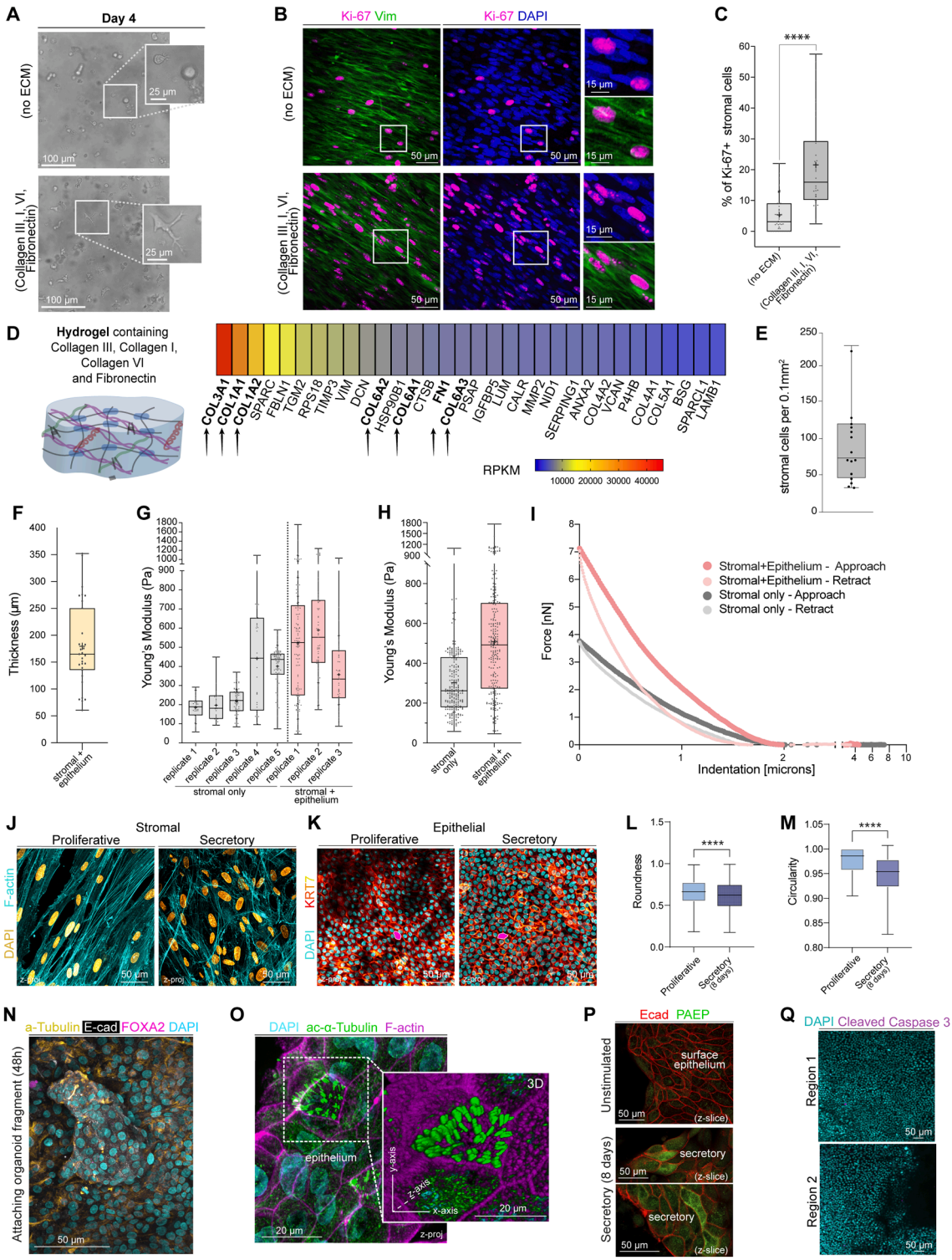

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Figure S1. Assembly and characterization of endometrial scaffold and cells: biophysical properties and morphometric analysis, related to Figures 1 and 2

(A) Bright-field imaging of stromal cells reveals a difference in cell elongation after 4 days of scaffold culture with (lower) and without (upper) ECM factors. Stromal cells in scaffolds containing ECM factors appear branched and elongated, whereas those in the no-ECM factor condition show less elongation. Scale bars, 100 μm.

(B) An increase of proliferation measured by the expression of Ki-67 protein was observed in stromal cells in the presence of ECM factors. Scale bar, 50 μm.

(C) Quantification of proliferative cells revealed a significant increase in the proliferative capacity of stromal cells when cultured with ECM factors, compared with the no-ECM factor condition (Mann-Whitney Test, ****$p < 0.0001$). Data are based on three individual experiments.

(D) Selection of ECM factors to incorporate into the hydrogel by identification of highly expressed ECM genes in endometrial biopsies (bulk RNA sequencing [RNA-seq]) that polymerize into fibrillary structures. Left, 3D schematic of the hydrogel.

(E) Immunofluorescence microscopy for VIM was used to measure stromal cell density. Each circle represents the data for one scaffold. Fifteen scaffolds were measured in three independent experiments.

(F) The thickness of the endometrial scaffold was measured using an atomic force microscope (AFM) ($n = 31$ measurements from 3 individual experiments).

(G) Stiffness measurements of individual endometrial scaffolds. The stiffness of the stromal compartment was assessed 24 h after seeding using AFM to determine the Young's modulus (5 samples were measured, with the number of measurement points per sample ranging from 28 to 86). Similarly, the stiffness of the complete endometrial scaffold, which includes both a stromal compartment and an outer luminal epithelial layer, was tested (3 samples were measured, with the number of measurement points per sample ranging from 49 to 139).

(H) Combined stiffness measurement for all samples.

(I) Representative force-displacement curves were measured by indentation using AFM for the stromal compartment (grey) and the full 3D endometrial scaffold (red). The bold lines represent the approach phase of the curve, while the lighter lines indicate the retraction phase.

(J) Morphology of endometrial stromal cells at proliferative and secretory phases. Images are representative of 3 independent experiments. Scale bars, 50 μm.

(K) Morphology of endometrial epithelial cells at proliferative and secretory phases. Images are representative of 3 independent experiments. Scale bars, 50 μm.

(L) Quantification of epithelial cell roundness $\left(4 \times area \,/\, \left[\pi \times major_axis^2\right]\right)$ in proliferative and secretory epithelial cells. Mann-Whitney test; ****, $p < 0.0001$, (number of cells, $n = 1{,}704$ (proliferative), $n = 826$ (secretory), 3 independent experiments). Outliers were identified and removed using the ROUT method (Q = 5%). Roundness ~ 0 indicates elongated morphology, and roundness ~ 1 indicates perfect circle.

(M) Quantification of circularity $\left(4\pi \times area \,/ perimeter^2\right)$ in proliferative and secretory epithelial cells. Mann-Whitney Test, ****, $p < 0.0001$ (number of cells, $n = 1{,}704$ [proliferative], $n = 826$ [secretory], 3 independent experiments). Outliers were identified and removed using the ROUT method (Q = 5%). Circularity measures shape roughness: cells that are round and smooth have a circularity of ~ 1, whereas cells with a rough perimeter and irregular borders have a circularity closer to 0.

(N) Top view of organoid fragments 48-h after seeding onto the surface of the stromal/hydrogel scaffold. Scale bars, 50 μm.

(O) Appearance of cilia on the surface of the endometrial epithelium after hormonal stimulation (E2, P4, and cAMP; 8 days): a subset of cells displays a single cilium, a subset of cells displays multiple cilia, and most of the cells do not display cilia. Scale bar, 20 μm.

(P) Immunofluorescence analysis of the epithelium before hormonal stimulation compared with the secretory phase (E2, P4, and cAMP; 8 days), showing a subset of epithelial secretory cells producing glycodelin. Scale bars, 50 μm.

(Q) Immunofluorescence microscopy of hormone-stimulated endometrial cells within scaffolds that have been cultured for 21 days. Very few cells have a signal for cleaved caspase-3, indicating a high level of viability in the cultures. Images from two scaffolds are shown. Scale bars, 50 μm.

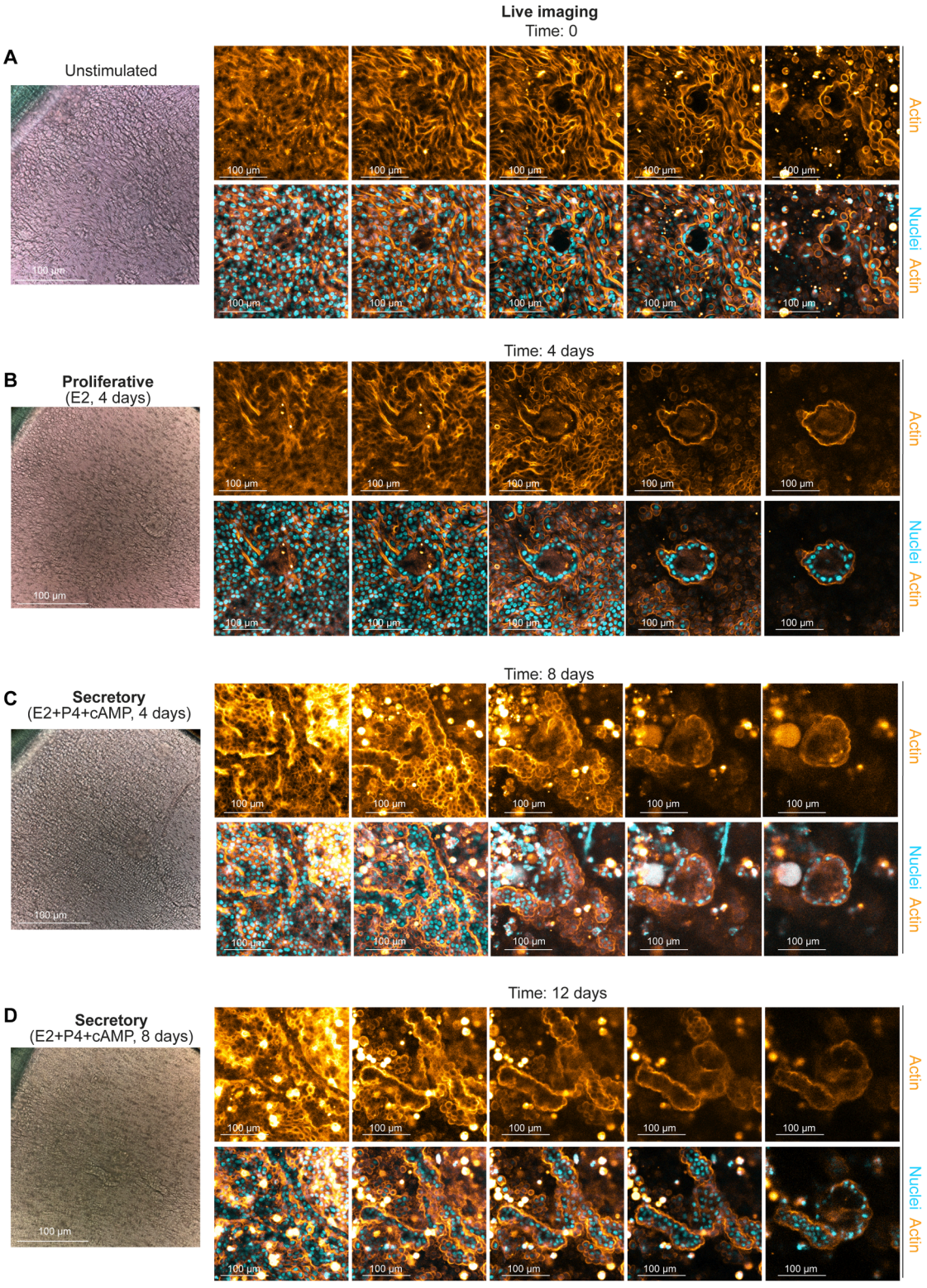

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Figure S2. Morphogenesis of epithelial glandular cells, related to Figure 2

(A–D) Images from live cultures following the same region of an endometrial scaffold at different stages of hormonal stimulation, corresponding to (A) unstimulated; (B) proliferative; (C) secretory, 4 days (totaling 8 days of culture); and (D) secretory, 8 days (totaling 12 days of culture). Left panels show bright-field views of the culture dish. The right panels show labeling of actin and nuclei with fluorescent dyes. Panels in a row show a series of z-sections.
All scale bars, 100 μm.

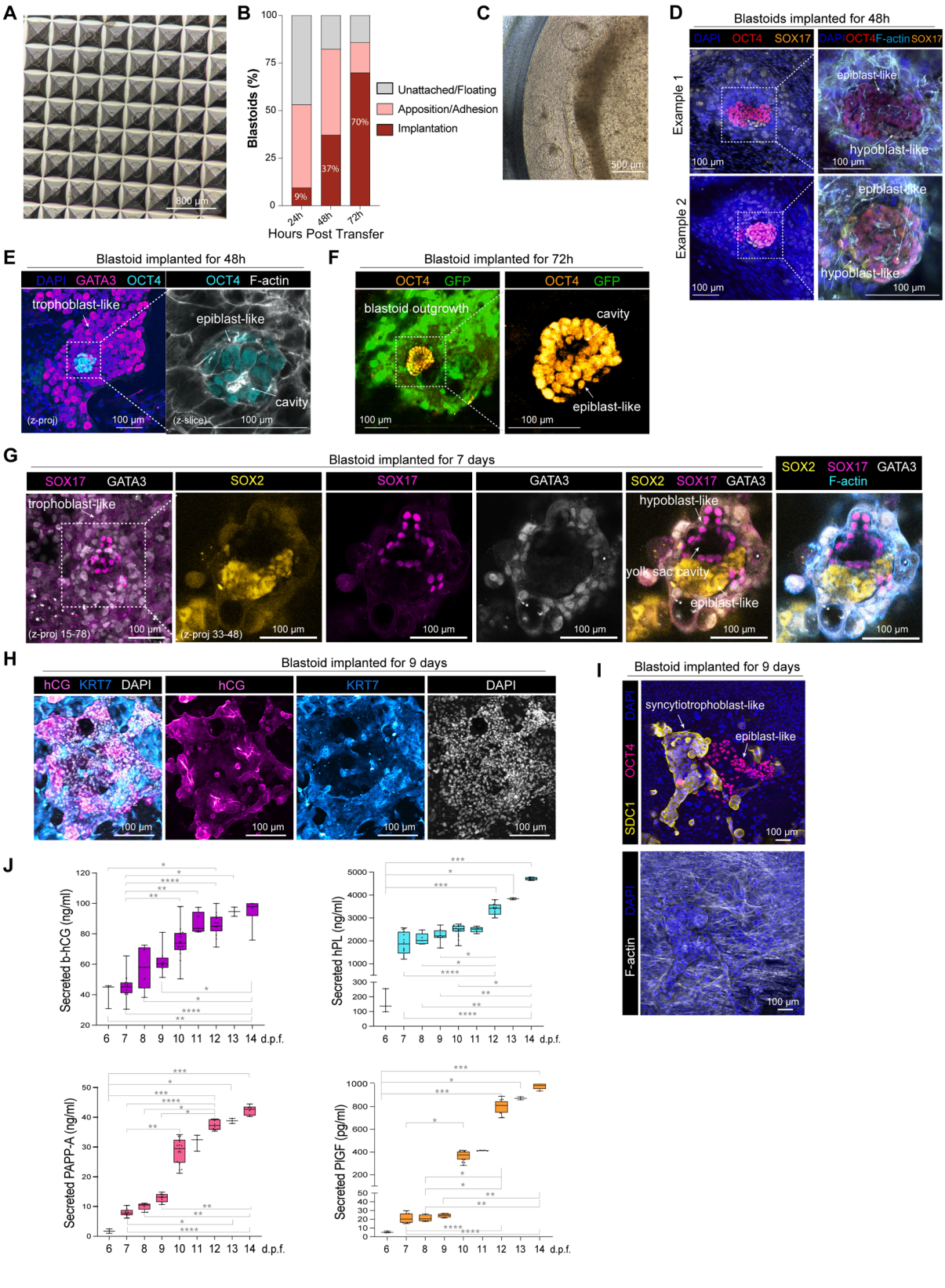

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Figure S3. Culture and implantation of human blastoids in endometrial scaffolds, related to Figure 3

(A) Blastoids in a well of an AggreWell plate, visualized via bright field.

(B) Quantification over time (24, 48, and 72 h culture) of percentages of blastoids unattached or attached to the endometrial scaffold (data from 7 independent experiments; total number of blastoids $n = 66$).

(C) Blastoids seeded onto CREST to implant, visualized with bright-field microscopy.

(D–I) Examples of implanted GFP-expressing human blastoids between 2 and 9 days of culture, with key cell types and features labeled. The site of implantation can be visualized by GFP. The epiblast-like cells are marked by OCT4, the hypoblast-like and YS-like cells are marked by SOX17-positive cells, cavities can be visualized by the enrichment of F-actin (representative of 1 blastoid out of 1 blastoid examined), and multi-nucleated STB-like cells are marked by hCG (representative of 4 blastoids out of 4 blastoids examined) and SDC1 (representative of 1 blastoid out of 1 blastoid examined). All scale bars, 100 μm.

(J) Dynamics and levels of hCG, hPL, PAPP-A, and PIGF secretion by implanted embryos into the culture medium during development within the endometrial scaffold. Values were collected from individually cultured embryos ($n = 3$ embryos, 6 d.p.f., $n = 15$ embryos, 7 d.p.f., $n = 5$ embryos, 8 d.p.f., $n = 8$ embryos, 9 d.p.f., $n = 19$ embryos, 10 d.p.f., $n = 5$ embryos, 11 d.p.f., $n = 13$ embryos, 12 d.p.f.; $n = 2$ embryos, 13 d.p.f., $n = 6$ embryos, 14 d.p.f.). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, derived from the Kruskal-Wallis test corrected by Dunn's.

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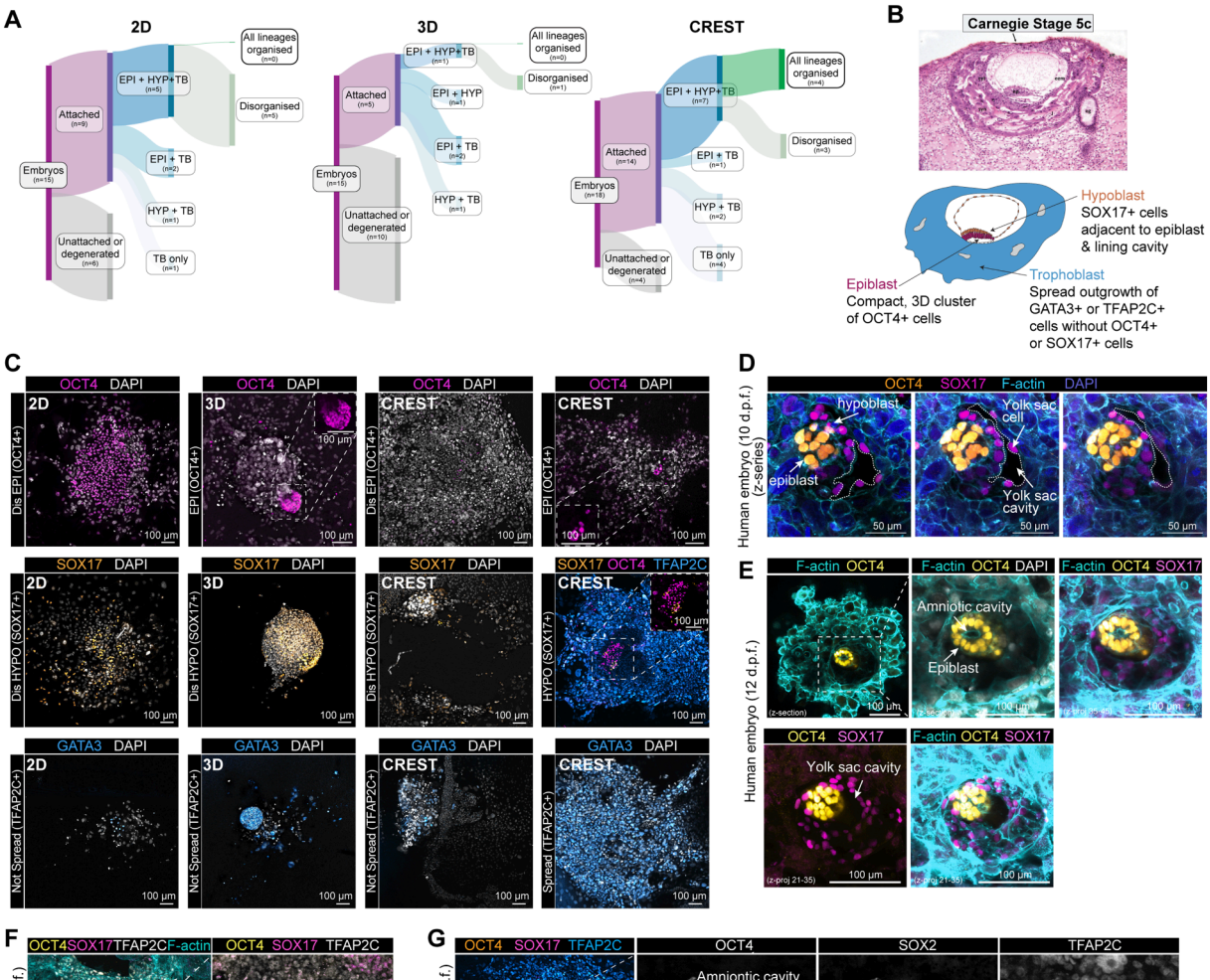

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Figure S4. Presence and organization of primary lineages in human embryos cultured in different systems and implanted in endometrial scaffolds, related to Figures 4 and 5

(A) Quantification of embryo implantation outcomes across three culture conditions (2D, 3D, and CREST), with the presence and organization of epiblast, hypoblast, and trophoblast lineages assessed at 14 d.p.f.

(B) Upper: image of an *in utero* CS5c post-implantation human embryo from the Carnegie collection, reproduced with kind permission from the Virtual Human Embryo project. Lower: schematic illustrating the criteria used for classifying organized epiblast, hypoblast, and trophoblast lineages in Figure 4.

(C) Examples of embryos with organized and disorganized epiblast, hypoblast, and trophoblast lineages that were cultured in either the 2D, 3D, or CREST conditions. Scale bar, 100 μm.

(D) Three z-sections of implanted human embryo at 10 d.p.f.: the epiblast is marked by OCT4+, the hypoblast is marked by columnar and highly packed SOX17+ cells lining the basal domain of the epiblast, and the yolk sac cavity is marked by flat and dispersed SOX17+ cells. Scale bars, 50 μm.

(E and F) Visualization of epiblast, hypoblast/visceral endoderm (VE), and YS lineages, and amniotic and yolk sac cavities in 14 d.p.f. embryos. Scale bars, 100 μm.

(G) Visualization of putative amniotic cells (TFAP2C+ OCT4+ SOX2–) in an *in vitro*-implanted human embryo at 14 d.p.f. Scale bars, 100 μm. Representative of 2 embryos out of 2 embryos examined.

(H and I) Images of two implanted embryos showing VIM-positive cells (cytoplasmic signal) that mark the emergence of extraembryonic mesoderm overlaying the hypoblast and lining the yolk sac cavity.

(J) Images showing the sequential, two-step labeling and image acquisition of an implanted embryo at 14 d.p.f. that built the composite image shown in Figure 5F. Left, the first round of imaging captured OCT4 to show the epiblast lineage, SOX17 to confirm the presence of hypoblast, and F-actin for a general assessment of the organization of the implantation site. Right, the second round of imaging captured GATA3 to show the extent of the trophoblast lineage, as well as VIM (cytoplasmic label). Scale bars, 100 μm.

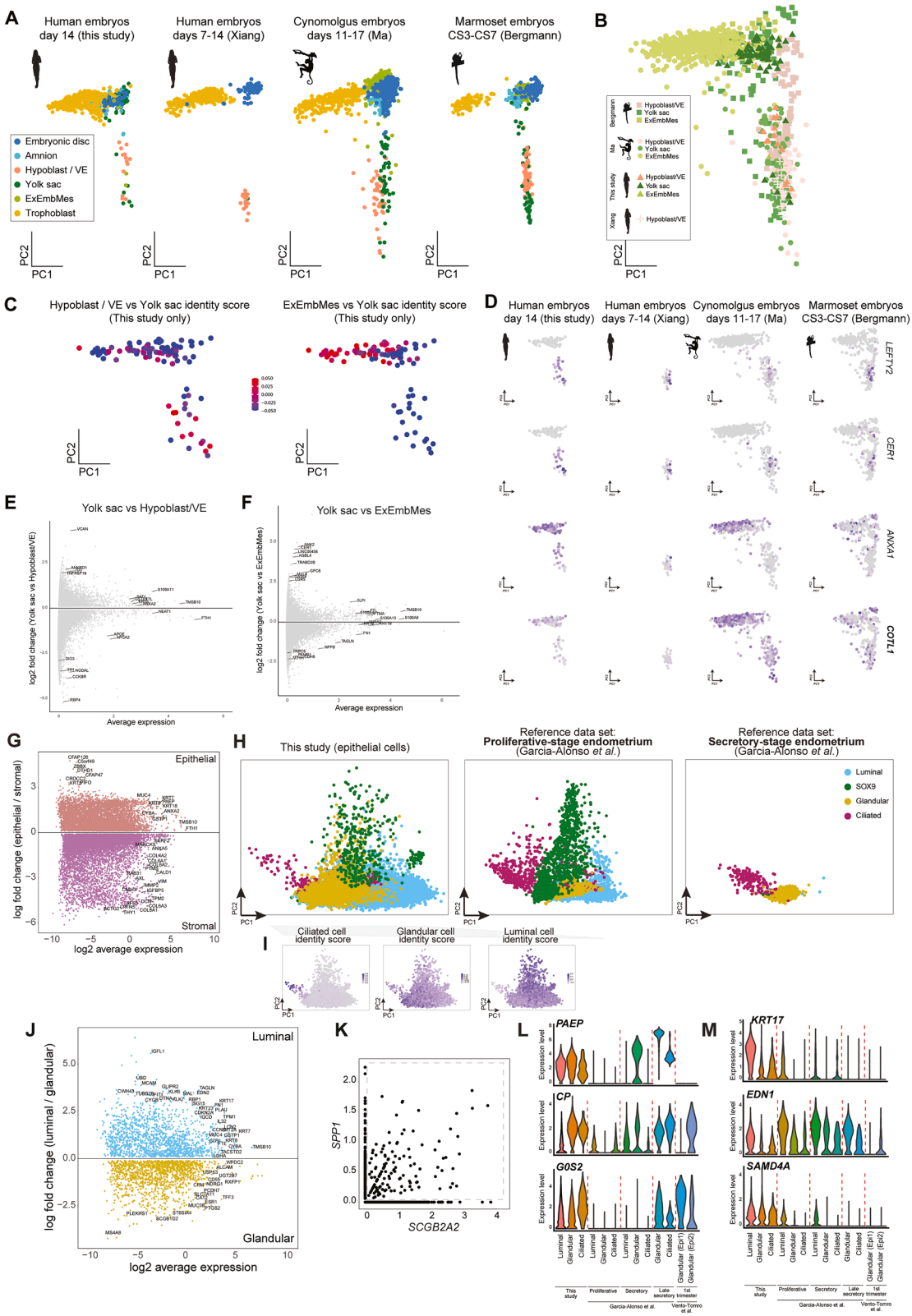

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Figure S5. Single-cell transcriptional profiling of endometrial epithelial cells within *in vitro* embryo implantation niche at 14 d.p.f., related to Figure 6

(A) PCA visualization following alignment of the embryonic and extraembryonic components of our dataset to human *in vitro*-cultured embryos, *in vitro*-cultured cynomolgus embryos, and marmoset embryos shows clear separation of indicated lineages.

(B) PCA of the presumed hypoblast-derived lineages compared with human, cynomolgus, and marmoset reference cells shows strong concordance of the hypoblast and presence of putative extraembryonic mesoderm.

(C) Confirmation of assigned cell identities. PCA of single-cell RNA-seq data from this study for presumed hypoblast-derived lineages. Each cell is shaded by an identity score for hypoblast/VE (left) cells and extraembryonic mesoderm cells based on differential expression in the cynomolgus dataset.

(D) Confirmation of hypoblast/VE and yolk sac cell identities by marker gene expression. PCA of single-cell RNA-seq data from this study (left-sided panels) and from CS5 and CS6 marmoset embryos[64] (right-sided panels) *LEFTY2* and *CER1* are markers of hypoblast/VE cells; *ANXA1* and *COTL1* are markers of yolk sac cells.

(E) MA plot showing genes differentially expressed between hypoblast/VE cells and yolk sac cells. Selected genes are labeled.

(F) MA plot showing genes differentially expressed between yolk sac cells and mesenchymal cells. Selected genes are labeled.

(G) MA plot of gene expression between endometrial stromal and epithelial cells. Selected genes are labeled.

(H) PCA of endometrial epithelial cells from this study (left panel) and from *in vivo* endometrial tissue biopsies[68] at proliferative (center panel) and secretory (right panel) phases. Datasets are aligned in Seurat, based on canonical correlation analysis.

(I) PCA of epithelial cells from this study colored by an identity score of the indicated cell type (ciliated, glandular, and luminal). Identity scores are based on the top 100 differentially expressed genes between the epithelial cell types in the Garcia-Alonso et al. dataset.[68]

(J) MA plot of gene expression between glandular and luminal epithelial cells. Selected genes are labeled.

(K) Further subpopulations could be identified according to expression of key markers, including glandular epithelia (*SCGB2A2* and *SPP1* double positive).

(L and M) Violin plots show expression levels of selected genes associated with either (L) glandular or (M) luminal identities in data from this study or from endometrial biopsy tissues at proliferative or secretary phases[68] or from two glandular populations of the decidua at 6–14 p.c.w.[67]

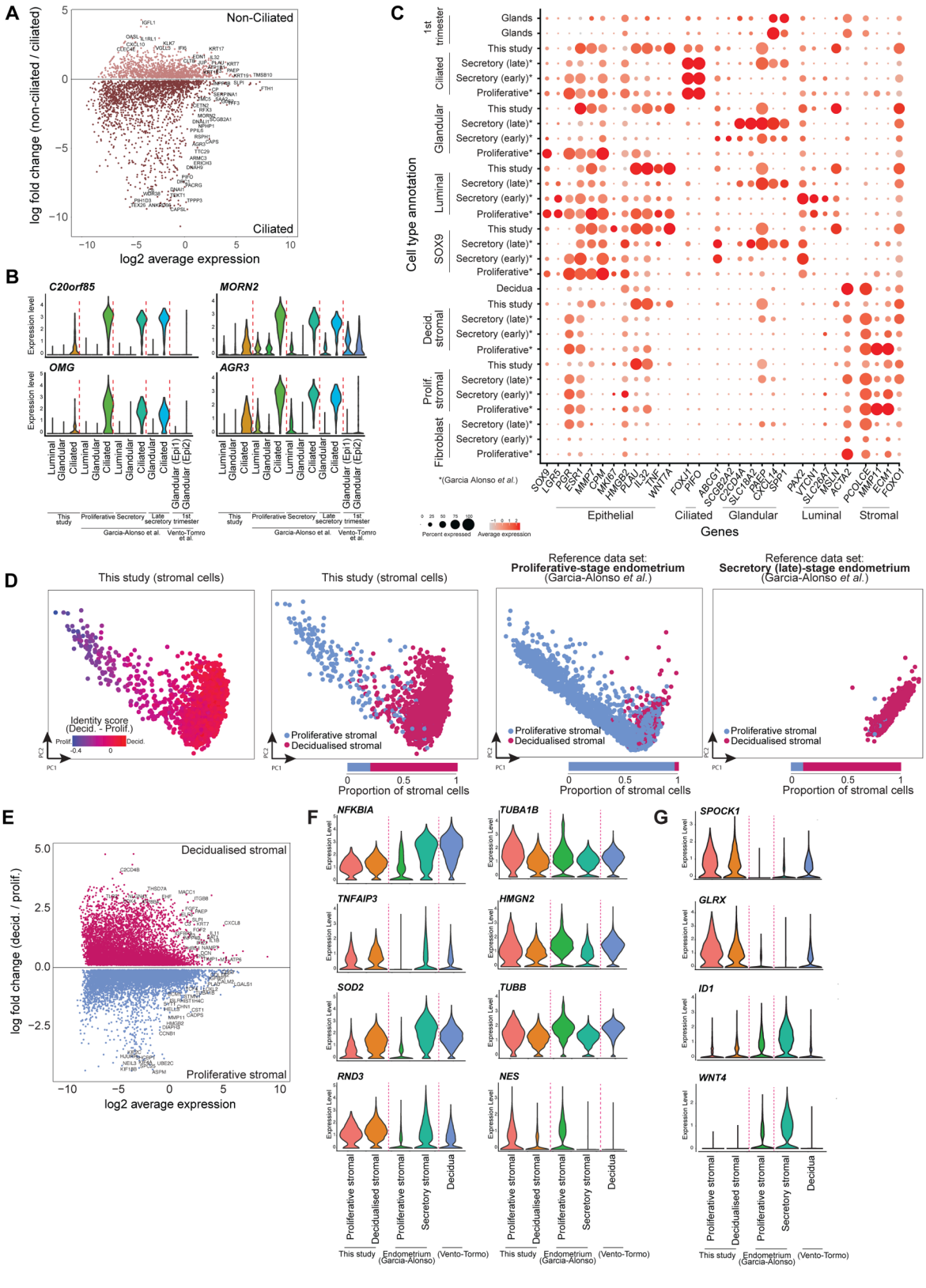

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Figure S6. Single-cell transcriptional profiling of endometrial epithelial and stromal cells within *in vitro* embryo implantation niche at 14 d.p.f., related to Figure 6

(A) MA plot of gene expression between ciliated and non-ciliated epithelial cells, based on model-based analysis of single-cell transcriptomics (MAST). Selected genes are labeled.

(B) Violin plots show expression levels of selected genes associated with ciliated epithelial identity in data from this study or from endometrial biopsy tissues at proliferative or secretary phases[68] or from two glandular populations of the decidua at 6–14 p.c.w.[67]

(C) Dotplot representation of gene expression of annotated epithelial and stromal cell subtypes in data from this study and from endometrial biopsy tissues at proliferative or secretary phases.[68]

(D) PCA of endometrial stromal cells from this study (left two panels) and from *in vivo* endometrial tissue biopsies[68] at proliferative and secretory phases, following alignment with Seurat. The left-hand panel colors each stromal cell with an identity score that is scaled from proliferative to decidualized; this identity score is derived by the top 100 differentially expressed genes between proliferative and decidualized cells in endometrial biopsies.[68] The three other panels show stromal cells colored by their closest identity to either proliferative or decidualized cells. The chart underneath each panel quantifies the proportion of cells in each stromal population assigned with either proliferative or decidualized identities. Most stromal cells in the 14 d.p.f. implantation site (this study) are assigned a transcriptional identity of decidualized stromal cells.

(E) MA plot of differentially expressed (DE) genes between decidualized and proliferative stromal cells. DE genes are identified using MAST within Seurat. Selected genes are labeled.

(F) Violin plots show expression levels of selected genes associated with either decidualized or proliferative identities in data from this study or from endometrial biopsy tissues at proliferative or secretary phases[68] or from the spongiosa or compacta regions of the decidua at 6–14 p.c.w.

(G) Violin plots show expression levels of a subset of genes that are expressed in first-trimester decidua,[67] including known decidual stress defense genes, such as *GLRX*, and that are also expressed in the endometrial stromal cells of our model at day 14 of embryo development but are not expressed in proliferative or secretory stromal cells of *in vivo* non-pregnant endometrial samples.[68] Additionally, genes known to influence the differentiation and proliferation of endometrial stromal cells, such as *ID1* and *WNT4*, which are expressed in stromal cells sampled from endometrial biopsies, are not expressed in first-trimester decidua nor in the endometrial stromal cells of our model at day 14 of embryo development. These data indicate that although the endometrial cells in our model originated from endometrial biopsies and are transcriptionally very close to their *in vivo* counterparts, the cells also present some distinct transcriptional responses that are associated with decidualized cells in pregnancy.

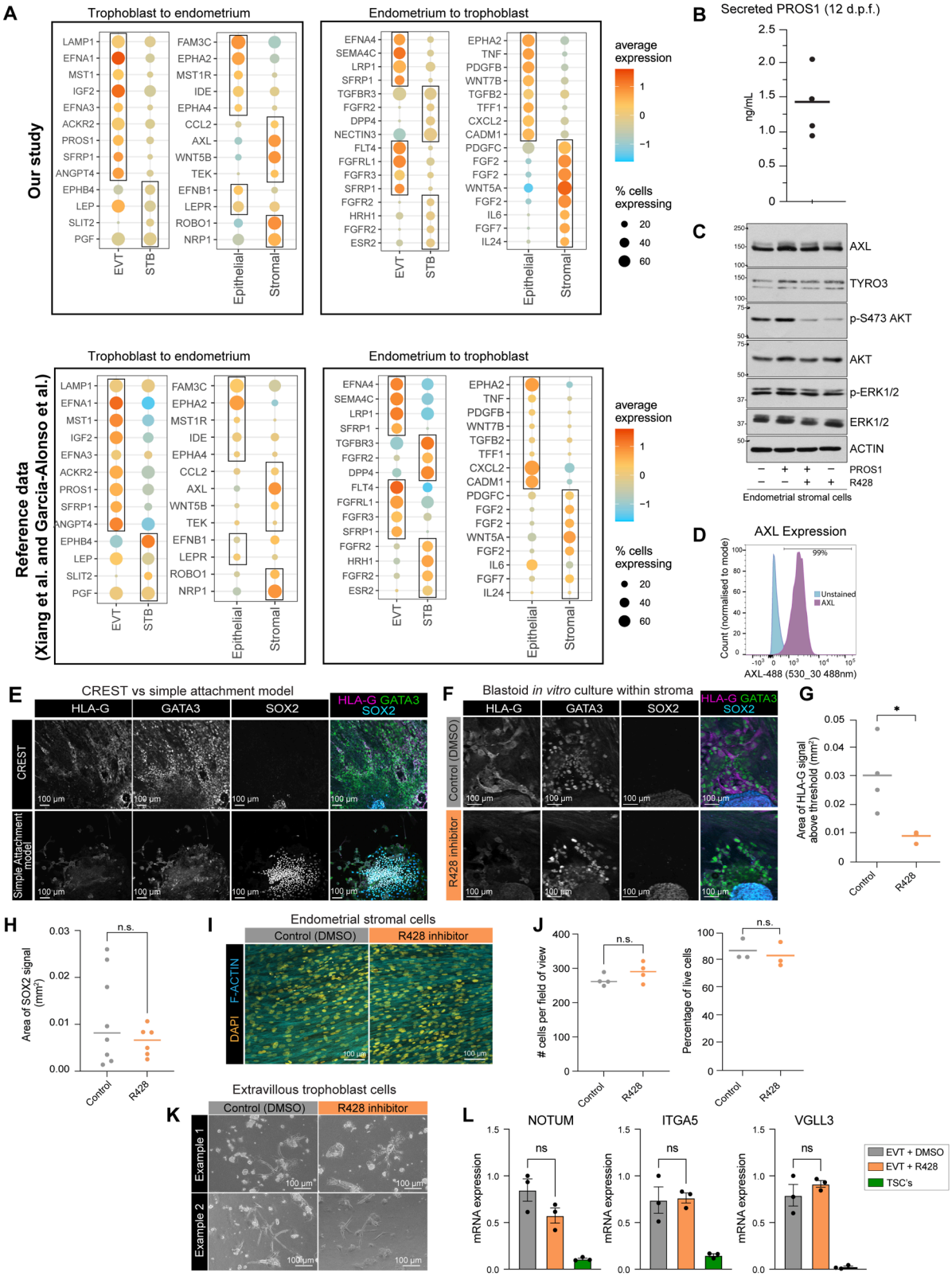

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Figure S7. Inhibition of PROS1-AXL in post-implantation blastoids impairs trophoblast outgrowth within the stromal compartment, related to Figure 7

(A) Expression of selected ligands and receptors in endometrial and trophoblast cell types. The upper panels show data from our study where the embryo was implanted *in vitro* within the endometrial cells. The lower panels show data from human embryos cultured *in vitro* in the absence of endometrial cells[28] and of cells from an endometrial biopsy[68].

(B) Quantification by ELISA of levels of PROS1 secreted into the culture medium. Measurements were taken on day 12 of human embryo implantation from four independent cultures. Mean and individual data points are shown.

(C) Endometrial stromal cells were treated with 1 μg/mL recombinant PROS1 and/or 500 nM R428 for 15 m. Whole cell lysates were fractionated by SDS-PAGE before immunoblotting with the indicated antibodies. Results are shown from a single experiment, and three experiments gave identical results. Molecular masses in kDa are indicated on the left-hand side.

(D) Flow cytometry analysis of AXL protein in endometrial stromal cells.

(E) Immunofluorescence microscopy images of embryos cultured *in vitro* in our endometrial model (CREST; upper) or in a simple attachment model lacking endometrial cells (lower). The images confirm the limited growth of trophoblast cells in the simple attachment model.

(F) Immunofluorescence microscopy images of *in vitro* blastoid implantation sites in control culture conditions (upper) and following 4 days of treatment with R428 inhibitor (lower). Representative images of 2 independent experiments.

(G) Quantification of area of HLA-G signal above threshold (left) and quantification of blastoid outgrowth (right) in individual implantation sites ($n = 4$ implanted blastoids for control; $n = 4$ implanted blastoids for R428 inhibitor). p values were derived from an unpaired, two-tailed t test; *, $p < 0.05$.

(H) Quantification of epiblast area (SOX2+ cells) ($n = 8$ embryos for control; $n = 6$ embryos for R428 inhibitor). p values were derived from an unpaired, two-tailed t test. One datapoint in the control group was removed following a Q-test for outlier identification.

(I) Images of endometrial stromal cells in control culture conditions (left) and following treatment for 4 days with the R428 inhibitor (right).

(J) Stromal cell growth is unaffected by the R428 inhibitor. Left, stromal cell number measured by immunofluorescent microscopy ($n = 4$ regions measured for control; $n = 4$ regions measured per condition; $p > 0.05$ from an unpaired, two-tailed t test); right, percentage of live stromal cells measured by flow cytometry ($n = 3$ independent samples per condition; $p > 0.05$ from an unpaired, two-tailed t test).

(K) Images show that the growth of EVT cells on their own is unaffected when cultured in the presence of the R428 inhibitor (right), as compared with control conditions (left). Images show two independent cultures.

(L) Gene-expression analysis by qPCR of BTS-11 trophoblast stem cells (TSCs) differentiated into EVT for 5 days and then treated with R428 or DMSO from days 5 to 7. Bars indicate the mean ± SEM (standard error of the mean) of three independent experiments shown as dots. Expression was normalized to EVT DMSO replicate 1.

All scale bars, 100 μm.